

Three-Piece Siege on the 4×4 Board is a Draw: A Certificate-Based Proof via Safety Games and a Colour-Reversible Rule Book

Kuangjiu Wang

23 August 2026

Abstract

Three-Piece Siege is a 4×4 board game with mandatory line captures, loss by reduction to two men or by immobility, and a draw by threefold repetition of the full position. We prove a draw without constructing a value table. The proof gives a set S of 91,330 positions that satisfies the local safety conditions for Black and contains the initial position with either side to move. A 180° rotation combined with colour and turn exchange transfers this certificate to White. Thus each player has a positional non-losing strategy; termination follows from the repetition rule. Verification is one pass of local tests over 3,723,720 positions, with no search or fixed-point iteration.

The certificate yields a 970-rule memoryless book over 79 Black shapes. Its Black-strategy reachable set Σ has 30,245 positions, is 67% smaller than S , and is itself a valid certificate; White uses the conjugate book. An auxiliary history-free retrograde classification is included only as corroboration and does not claim to solve the history-sensitive game exactly.

The precise claim is that, under the history-dependent threefold-repetition rule, each player has a positional strategy guaranteeing at least a draw. Lemma 2.7 guarantees termination; Theorem 3.2, not finiteness alone, establishes the effectiveness of the safety strategies. The verification is computer-assisted rather than machine-formalised: the published C verifier and Python rule-book checker check the certificates, and no proof-assistant formalisation has been attempted.

1 Introduction

Retrograde analysis turns finite-game reasoning into an algorithm [4, 3], but a computed value is not by itself an independently checkable proof. The distinction, familiar from certifying algorithms [2], is between an algorithm that outputs an answer and one that outputs a *witness*. Ströhlein’s early formulation and Romein–Bal’s Awari solver provide the direct retrograde-analysis context; the present work uses such a classification only as an auxiliary comparison, not as its main proof.

For a drawing claim, a clean witness is a set of positions closed under the relevant moves and avoiding losing states. These are local conditions: each concerns one position and its immediate successors. Such sets are safety-game invariants and greatest fixed points of monotone operators [5, 1]; they provide finite, checkable certificates for non-losing strategies.

We apply this certificate method to *Three-Piece Siege*, a 4×4 game whose formal rules are given in Section 2. Mandatory geometry-triggered captures, loss at two men, and history-sensitive threefold repetition make it a compact but nontrivial test case. We make no claim of new safety-game theory: the contribution is an explicit certificate and verification for this rule set. More precisely, this is a verifiable certificate-based computational proof case study: it

supplies an auditable artifact and independent local checks for one rule set, rather than a new general safety-game theorem or merely an informal analysis of a board game.

Our results are the following.

- The game is a draw (Theorem 5.2). The proof needs a single explicitly given set S of 91,330 positions, two local closure properties of S , and one symmetry of the rules. It does not use the values of any position.
- The history-dependence of the draw rule is harmless (Lemma 2.7): because each full position can occur at most three times, every play has at most $2N + 1$ moves with $N = 3,723,720$, so the game tree is finite. The position-only non-loss claims are justified separately by the safety-certificate theorem (Theorem 3.2). This settles a caveat raised in the rules document itself.
- There is a memoryless rule book Π of 970 rules, indexed by the 79 shapes that Black’s men may form. Black plays Π directly, while White plays the conjugate book Π^p under the colour-reversing symmetry (Theorem 7.2). The reachable set for Π , of only 30,245 positions, is again a valid certificate (Theorem 7.3), so the strategy and the proof are the same object.
- The certificate has a transparent structure (Propositions 5.5–5.7): all of Black’s shapes live in a fixed 10-square region; Black concedes at most one man in the whole game; and once Black is down to three men, White never again has a capturing move at all — in none of the 8,968 relevant positions.
- An auxiliary history-free retrograde classification (Section 8) is used only as corroboration, not as an exact solution of the history-sensitive game. It yields the local structural fact (Proposition 8.1) that, of the 1,820 four-man shapes, exactly 1 — the central 2×2 block — is safe against every opposing configuration.

The results are certificate-based rather than value-table based: a 232-line verifier checks two closure conditions on the exhibited set, and the colour-reversing symmetry transfers the certificate from one player to the other. An auxiliary history-free retrograde classification is reported only for comparison. The contribution is therefore a small, independently checkable computational case study.

2 The game

We follow the normative rules document (version 2.0, 22 August 2026); section numbers in the form “§7.2” refer to it. The definitions below are a verbatim formalisation.

2.1 Board, positions, moves

Let $X = \{1, 2, 3, 4\}^2$, whose elements (*squares*) we write rc with r the row (from the top) and c the column (from the left); thus 41 is the bottom-left square. Let d_1 be the ℓ^1 distance on X . A *line* is a row $R_r = \{r\} \times \{1, \dots, 4\}$ or a column $C_c = \{1, \dots, 4\} \times \{c\}$; there are eight lines. Black and White each own a set of indistinguishable men.

Definition 2.1 (positions). A *position* is a triple $s = (B, W, t)$ with $B, W \subseteq X$, $B \cap W = \emptyset$, $|B|, |W| \in \{3, 4\}$ and $t \in \{b, w\}$; t is the side to move. We write $\mathcal{S}$ for the set of positions and $N = |\mathcal{S}|$. The *mover’s set* is $P(s) = B$ if $t = b$ and $P(s) = W$ if $t = w$; the *opponent’s set* $Q(s)$ is the other one.

Positions are exactly the states from which play can continue (§5.1): a player reduced to two men has already lost, so $|B|, |W| \geq 3$. Counting,

$$N = 2 \left[\binom{16}{4} \binom{12}{4} + \binom{16}{4} \binom{12}{3} + \binom{16}{3} \binom{13}{4} + \binom{16}{3} \binom{13}{3} \right] = 3,723,720.$$

The *initial position* has Black on the bottom row and White on the top row, with Black to move:

$$B_0 = \{41, 42, 43, 44\}, \quad W_0 = \{11, 12, 13, 14\}, \quad s_0^b = (B_0, W_0, b),$$

and we also write $s_0^w = (B_0, W_0, w)$ for the same board with White to move.

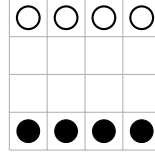


Figure 1: The initial position s_0^b (§3). Black (•) is to move.

Definition 2.2 (plain moves). The set of *plain moves* at s is $\mathcal{M}(s) = \{(x, y) : x \in P(s), y \notin B \cup W, d_1(x, y) = 1\}$ (§6). For $m = (x, y) \in \mathcal{M}(s)$ put $P' = (P(s) \setminus \{x\}) \cup \{y\}$.

Captures are not declared by the player; they are detected automatically after the move, and only on the two lines through the destination square (§7.1).

Definition 2.3 (capture candidates). Let $m = (x, y) \in \mathcal{M}(s)$, $Q = Q(s)$ and let L be one of the two lines containing y . A square z is a *capture candidate on L* if, writing $O_L = (P' \cup Q) \cap L$ (§7.2),

- (i) $|O_L| = 3$; (ii) $|P' \cap L| = 2$; (iii) $Q \cap L = \{z\}$;
- (iv) writing $P' \cap L = \{a, b\}$, one has $d_1(a, b) = 1$;
- (v) $d_1(z, a) = 1$ or $d_1(z, b) = 1$.

Write $\mathcal{Z}(s, m)$ for the set of capture candidates on either of the two lines through y .

Thus a capture occurs precisely when the move creates, on a line through the destination, a configuration of exactly three men consisting of two adjacent men of the mover and one adjacent man of the opponent — a sandwich seen from one side, whence the name. Since $y \in P' \cap L$, the moved man is automatically one of a, b .

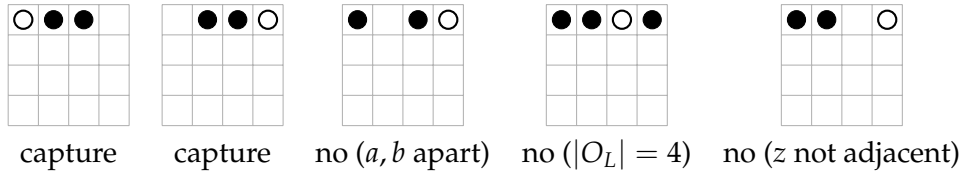


Figure 2: Line patterns for Black to move, after Black's man has arrived on the line (§7.3–7.4). Only the row and the column through the destination square are examined.

Definition 2.4 (actions). The set of *actions* at s is

$$\mathcal{A}(s) = \{(m, z) : m \in \mathcal{M}(s), z \in \mathcal{Z}(s, m)\} \cup \{(m, \perp) : m \in \mathcal{M}(s), \mathcal{Z}(s, m) = \emptyset\}.$$

Capturing is mandatory (§8): if $\mathcal{Z}(s, m) \neq \emptyset$ the mover *must* remove one candidate, and if there are two candidates the mover chooses which (§8.3). At most one man is captured per turn and chain captures are forbidden (§8.4–8.5). Since every plain move yields at least one action, we record:

$$\mathcal{A}(s) = \emptyset \iff \mathcal{M}(s) = \emptyset. \quad (1)$$

Definition 2.5 (transition). Let $a = (m, z) \in \mathcal{A}(s)$ with $m = (x, y)$. Put $Q' = Q(s) \setminus \{z\}$ if $z \neq \perp$ and $Q' = Q(s)$ otherwise. If $|Q'| = 2$ we call a *winning* and write $s \cdot a = \top$. Otherwise $s \cdot a \in \mathcal{S}$ is the position with mover's set P' , opponent's set Q' and the other side to move.

2.2 Plays, termination, and the repetition rule

Definition 2.6 (plays). A *play* from $s_0 \in \mathcal{S}$ is a finite sequence $\pi = (s_0, a_0, s_1, a_1, \dots)$ with $a_k \in \mathcal{A}(s_k)$ and $s_{k+1} = s_k \cdot a_k \in \mathcal{S}$, which ends in exactly one of the following three ways. Let $H_k = (s_0, \dots, s_k)$ and let $\text{occ}_{H_k}(s)$ be the number of indices $j \leq k$ with $s_j = s$. Immediately after the mover at s_k has chosen a_k , the following tests are applied *in this order* (§11.4):

(T1) if a_k is winning, the play ends and the mover at s_k **wins** (§8, §9);

(T2) otherwise $s_{k+1} = s_k \cdot a_k$ is appended, and if $\mathcal{M}(s_{k+1}) = \emptyset$ the play ends and the mover at s_{k+1} **loses** (§10);

(T3) otherwise, if $\text{occ}_{H_{k+1}}(s_{k+1}) = 3$ the play ends in a **draw** (§11.2).

If none of (T1)–(T3) applies, play continues from s_{k+1} . Test (T2) is also applied to s_0 itself before the first move.

Two features of this definition deserve emphasis. First, the object counted for repetition is the *full position* (B, W, t) , not the board pattern (§11.1), and the initial position counts as its own first occurrence (§11.2). Second, the priority order matters: a player who is immobilised loses even if the position is simultaneously a third repetition (§11.4).

Because termination depends on the history, a strategy is in general a function of histories, not of positions, and the rules document explicitly warns (§17) that an exact analysis must use the extended state (B, W, t, H) . The following elementary observation removes the difficulty for our purposes.

Lemma 2.7 (finiteness). *Every play from any $s_0 \in \mathcal{S}$ has at most $2N + 1$ moves, where $N = |\mathcal{S}| = 3,723,720$. In particular the game is finite, and every maximal play ends by one of (T1), (T2), (T3).*

Proof. Say that a position is *appended* when it is added to the history; s_0 counts as appended once, and each move either ends the play by (T1) or appends exactly one position. Hence, if the play has ℓ moves, the number of appendings is ℓ or $\ell + 1$; in either case ℓ does not exceed the number of appendings.

Suppose some position reaches three appendings, say s_{k+1} with $\text{occ}_{H_{k+1}}(s_{k+1}) = 3$. Then the play ends at that point: (T1) has already been passed, so if (T2) does not fire, (T3) does. Consequently at most one position is appended three times and it is the last appending, while every other position is appended at most twice. The number of appendings is therefore at most $2(N - 1) + 3 = 2N + 1$, and so is the number of moves.

Finally, if a play has not ended at s_k then $\mathcal{M}(s_k) \neq \emptyset$ by (T2), hence $\mathcal{A}(s_k) \neq \emptyset$ by (1) and the mover has an action available. So a maximal play cannot go on forever and cannot get stuck without a verdict: it ends, and only by (T1)–(T3). $\square$

Remark 2.8. The bound $2N + 1$ is of course absurdly pessimistic — the longest line observed in the auxiliary history-free analysis is shorter than 30 moves — but its only role is to guarantee that no play is infinite. It does not by itself imply that positional strategies suffice for

a history-dependent game. For the strategies constructed below, that conclusion comes from Theorem 3.2: the repetition rule can only ever *add* a draw termination, never a loss, so a strategy certified not to lose on any finite prefix never loses at all.

3 Safety certificates

Throughout, $P \in \{\text{Black}, \text{White}\}$ is a distinguished player and $\bar{P}$ is the opponent. We say a position s is *P-to-move* if $t = b$ and P is Black, or $t = w$ and P is White.

Definition 3.1 (*P-safe set*). A set $S \subseteq \mathcal{S}$ is *P-safe* if

- (S1) for every *P-to-move* $s \in S$ there exists $a \in \mathcal{A}(s)$ such that a is winning or $s \cdot a \in S$;
- (S2) for every $\bar{P}$ -to-move $s \in S$ and every $a \in \mathcal{A}(s)$: a is not winning, and $s \cdot a \in S$.

Both conditions are local: (S1) quantifies over the actions at a single position and (S2) likewise, and membership in S is a table lookup. Note that (S1) forces $\mathcal{A}(s) \neq \emptyset$, hence $\mathcal{M}(s) \neq \emptyset$ by (1), so a *P-to-move* position of a *P-safe* set is never immobile. Condition (S2) is vacuous at a $\bar{P}$ -to-move position with $\mathcal{A}(s) = \emptyset$; such a position is a win for P by (T2).

A *positional* (or memoryless) strategy for P is a map σ assigning to each *P-to-move* position s with $\mathcal{A}(s) \neq \emptyset$ an action $\sigma(s) \in \mathcal{A}(s)$; a play is *consistent with* σ if P 's every action is given by σ .

Theorem 3.2 (soundness). *Let S be P-safe and $s_0 \in S$. Then there is a positional strategy σ for P such that every maximal play from s_0 consistent with σ ends either with a win for P by (T1) or (T2), or in a draw by (T3). In particular $\bar{P}$ has no winning strategy from s_0 , and P has a strategy guaranteeing at least a draw.*

Proof. For each *P-to-move* $s \in S$ choose, using (S1), an action $\sigma(s) \in \mathcal{A}(s)$ that is winning or satisfies $s \cdot \sigma(s) \in S$ (a choice from finitely many finite sets); define σ arbitrarily outside S . Let π be a maximal play from s_0 consistent with σ .

We claim that every position appended in π lies in S , and that π does not end with a win for $\bar{P}$. We argue by induction along π ; the base case $s_0 \in S$ is the hypothesis. Let $s_k \in S$ and suppose the play has not yet ended at s_k .

If s_k is *P-to-move*, then $\mathcal{M}(s_k) \neq \emptyset$ as observed above, so the play was not ended by (T2) at s_k for want of a move. P plays $\sigma(s_k)$: either it is winning, and π ends with a win for P by (T1); or $s_{k+1} = s_k \cdot \sigma(s_k) \in S$ and the play either ends by (T2) — which is then a loss for the mover at s_{k+1} , i.e. for $\bar{P}$, a win for P — or by (T3) as a draw, or continues.

If s_k is $\bar{P}$ -to-move, then either $\mathcal{M}(s_k) = \emptyset$, in which case π already ended by (T2) with a loss for $\bar{P}$; or $\bar{P}$ plays some $a \in \mathcal{A}(s_k)$. By (S2) a is not winning, so π does not end by (T1) here, and $s_{k+1} = s_k \cdot a \in S$. As before the play may then end by (T2) (a loss for the mover at s_{k+1} , i.e. for P — but this is impossible, since $s_{k+1} \in S$ is *P-to-move* and hence not immobile), or by (T3) as a draw, or continue.

This proves the claim. By Lemma 2.7 the play π is finite, and being maximal it ends by one of (T1)–(T3); by the claim the only possibilities are a win for P or a draw. $\square$

Remark 3.3 (why a greatest fixed point). The union of two *P-safe* sets is *P-safe*, so there is a largest one, S_p^* ; it is the greatest fixed point of the monotone operator $F(S) = \{s : \text{(S1) or (S2) holds for } s \text{ w.r.t. } S\}$ and can be computed by iterated deletion from $\mathcal{S}$ (Tarski [5]; see [1] for safety games). S_p^* is the exact non-losing region for the history-free safety objective defined by (S1)–(S2). For the actual game, whose repetition rule is history-sensitive, it is a certified subset of that game's non-losing region. The point of Definition 3.1 is not the existence but the *form*: any S whatsoever, however obtained, proves the theorem once (S1) and (S2) are checked, and the check never mentions the iteration that produced S .

Remark 3.4 (complexity of checking). Verifying (S1)–(S2) for a given $S \subseteq \mathcal{S}$ costs one pass over $\mathcal{S}$, generating the actions of each position; with a bitboard representation of B and W this is $O(N)$ time and $O(N)$ space with tiny constants. Our verifier (Section 6) does it in 0.12 s.

4 The symmetries of the rules

Let

$$\mu(r, c) = (r, 5 - c), \quad \tau(r, c) = (5 - r, 5 - c).$$

Both are involutive bijections of X ; μ is the left–right mirror and τ the 180° rotation. We extend them to sets of squares elementwise, and define

$$\hat{\mu}(B, W, t) = (\mu B, \mu W, t), \quad \boxed{\rho(B, W, t) = (\tau W, \tau B, \bar{t})}$$

where $\bar{b} = w$ and $\bar{w} = b$. Thus ρ rotates the board, exchanges the colours, and exchanges the side to move.

Lemma 4.1. ρ is an involution of $\mathcal{S}$, and for every $s \in \mathcal{S}$ there is a bijection $\mathcal{A}(s) \rightarrow \mathcal{A}(\rho s)$, $a \mapsto \rho_s a$, such that

- (i) a is winning if and only if $\rho_s a$ is winning;
- (ii) if a is not winning then $\rho(s \cdot a) = \rho s \cdot \rho_s a$;
- (iii) $\mathcal{A}(s) = \emptyset$ if and only if $\mathcal{A}(\rho s) = \emptyset$.

The same holds with $\hat{\mu}$ in place of ρ .

Proof. That ρ is an involution of $\mathcal{S}$ is immediate: τ is a bijection preserving disjointness and cardinality, and $\bar{\bar{t}} = t$.

The mover’s set at s is $P(s)$, and the mover’s set at ρs is $\tau P(s)$: if $t = b$ then $P(s) = B$, and the side to move at ρs is w whose set is τB ; symmetrically for $t = w$. Likewise the opponent’s set at ρs is $\tau Q(s)$. So ρ maps the mover to the mover and the opponent to the opponent.

Now τ is an automorphism of the grid graph on X : it is a bijection with $d_1(\tau u, \tau v) = d_1(u, v)$. It also permutes the eight lines, since $\tau R_r = R_{5-r}$ and $\tau C_c = C_{5-c}$, and it maps the two lines through y onto the two lines through τy . Define $\rho_s(x, y) = (\tau x, \tau y)$ on plain moves; by the above this is a bijection $\mathcal{M}(s) \rightarrow \mathcal{M}(\rho s)$, and it maps P' to $(\tau P)'$. Conditions (i)–(v) of Definition 2.3 mention only cardinalities of intersections of P' and Q with a line and d_1 -adjacencies between the men on that line; all of these are preserved. Hence $\tau \mathcal{Z}(s, m) = \mathcal{Z}(\rho s, \rho_s m)$, and setting $\rho_s(m, z) = (\rho_s m, \tau z)$ (with $\tau \perp = \perp$) gives a bijection $\mathcal{A}(s) \rightarrow \mathcal{A}(\rho s)$. Since $|\tau Q'| = |Q'|$, an action is winning iff its image is, which is (i); and (ii) holds by construction, because the image position has mover’s set $\tau P'$, opponent’s set $\tau Q'$ and the opposite side to move, which is precisely $\rho(s \cdot a)$. Finally (iii) follows from (1) and the bijection on plain moves. The argument for $\hat{\mu}$ is identical, using that μ is a grid automorphism permuting the lines and that $\hat{\mu}$ preserves colours. $\square$

Corollary 4.2. If S is Black-safe then $\rho(S)$ is White-safe, and conversely. If S is P-safe then so is $\hat{\mu}(S)$.

Proof. Let S be Black-safe and put $S' = \rho(S)$; note ρ maps Black-to-move positions to White-to-move positions and conversely. Let $s' = \rho s \in S'$ be White-to-move, so $s \in S$ is Black-to-move; by (S1) there is $a \in \mathcal{A}(s)$ winning or with $s \cdot a \in S$, and then $\rho_s a \in \mathcal{A}(s')$ is winning, resp. satisfies $s' \cdot \rho_s a = \rho(s \cdot a) \in S'$, by Lemma 4.1. This is (S1) for S' with $P = \text{White}$. Let $s' = \rho s \in S'$ be Black-to-move and $a' \in \mathcal{A}(s')$; then $a' = \rho_s a$ for a unique $a \in \mathcal{A}(s)$ (using $\rho^2 = \text{id}$), a is not winning by (S2) for S , hence neither is a' , and $s' \cdot a' = \rho(s \cdot a) \in S'$. This is (S2). The converse and the $\hat{\mu}$ statement are proved the same way. $\square$

The following triviality is what couples the two colours.

Lemma 4.3. $\rho(s_0^w) = s_0^b$ and $\rho(s_0^b) = s_0^w$.

Proof. $\tau W_0 = \tau\{11, 12, 13, 14\} = \{44, 43, 42, 41\} = B_0$ and likewise $\tau B_0 = W_0$; and $\bar{w} = b$. Hence $\rho(B_0, W_0, w) = (\tau W_0, \tau B_0, b) = (B_0, W_0, b)$. $\square$

Remark 4.4. This is the precise sense in which Three-Piece Siege is a *symmetric* game: the two players face the same problem, but with the side to move swapped. Consequently a single certificate that contains the initial board *both* with Black to move and with White to move settles both colours at once. It is not a priori clear that such a certificate exists: it asserts that Black can survive even after conceding the first move.

5 The certificate

5.1 Statement

We exhibit a set $S \subseteq \mathcal{S}$ of size 91,330. It is given as a flat array of one-byte flags indexed by $(\text{rank}(B), \text{rank}(W), t)$, where rank enumerates the 2380 subsets of X of size 3 or 4 in increasing order of their 16-bit characteristic words (bit i set iff square i , in row-major order, is occupied); precisely, byte $(\text{rank}(B) \cdot 2380 + \text{rank}(W)) \cdot 2 + t$ of the file is 1 if $(B, W, t) \in S$ and 0 otherwise (this is the function `sidx` of `engine.py`). The file `safe6.bin` of $2380^2 \cdot 2 = 11\,328\,800$ bytes is part of the supplementary material; Section 9 gives its checksum and the environment needed to regenerate it. Its statistics are

$$\begin{aligned} |S| &= 91,330, \\ \text{Black to move: } 63,885, \quad &\text{White to move: } 27,445, \\ (|B|, |W|) &= (4, 4) : 26,646, \quad (4, 3) : 28,800, \\ (3, 4) &: 13,042, \quad (3, 3) : 22,842. \end{aligned}$$

Theorem 5.1. S is Black-safe, and $s_0^b \in S$ and $s_0^w \in S$.

The proof is a finite verification, described and audited in Section 6.

Theorem 5.2 (main). *Three-Piece Siege on the 4×4 board, with the rules of Section 2, is a draw: each player has a positional strategy that guarantees at least a draw. Moreover any maximal play in which one of these strategies is followed, and which is not won by the player following it, ends in a draw by threefold repetition after at most $2N + 1$ moves.*

Proof. By Theorem 5.1, S is Black-safe and $s_0^b \in S$; by Theorem 3.2 with $P = \text{Black}$, Black has a positional strategy guaranteeing at least a draw from s_0^b , and White has no winning strategy.

By Theorem 5.1 and Corollary 4.2, $S' = \rho(S)$ is White-safe. By Theorem 5.1 again, $s_0^w \in S$, so by Lemma 4.3 $s_0^b = \rho(s_0^w) \in S'$. By Theorem 3.2 with $P = \text{White}$, White has a positional strategy guaranteeing at least a draw from s_0^b , and Black has no winning strategy. The bound on the length of a play is Lemma 2.7. $\square$

Remark 5.3. No appeal to a full value-table or backward-induction computation is made: both non-losing strategies are exhibited, and “draw” means precisely that neither player has a winning strategy while both have non-losing ones. In particular the argument is unaffected by the history-dependence of (T3).

Remark 5.4 (the first-repetition variant). Under the variant of §13, in which the *second* occurrence of a full position is already a draw, Lemma 2.7 improves to $N + 1$ moves and nothing else changes: Definition 3.1 and Theorem 3.2 never mention how many repetitions are needed, only that a repetition draw is not a loss. Hence Theorem 5.2 holds verbatim for that variant, with the same certificate S and the same strategies.

5.2 Where S came from

The construction is heuristic and plays no part in the proof; we record it for reproducibility. Starting from the full state space we compute the greatest fixed point of the safety operator for Black by iterated deletion (program `safety6.c`: 24 sweeps, 0.19 s). This yields the largest Black-safe set, which is huge and structureless. To obtain something intelligible we restrict Black’s admissible shapes: fix the 10-square region

$$\mathcal{R} = \{22, 23\} \cup R_3 \cup R_4$$

and a whitelist $\mathcal{W}$ of subsets of $\mathcal{R}$ of size 3 or 4, and compute the greatest fixed point of the safety operator *restricted* to positions with $B \in \mathcal{W}$ — that is, Black must at every moment be able to keep his men inside the whitelist. The whitelist is then minimised greedily: shapes (in $\hat{\mu}$ -orbits, to keep the result mirror-symmetric) are proposed for deletion in order of decreasing distance from Black’s home block, and a deletion is accepted if the restricted fixed point still contains *both* s_0^b and s_0^w (program `prune3.py`, 170 orbit trials, each a full re-solve). The surviving whitelist has 153 shapes and the associated fixed point is S .

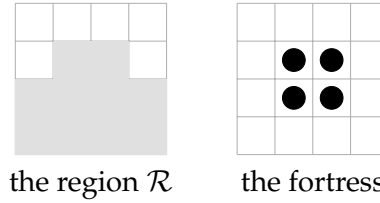


Figure 3: Left: the 10 squares of $\mathcal{R}$; Black’s men never leave this region in the certificate (Proposition 5.5). Right: the central 2×2 block, the unique four-man shape that is safe against every opposing configuration (Proposition 8.1).

5.3 Structure of S

The following propositions report exhaustive checks over S . They are not needed for Theorem 5.2, but they make the certificate readable.

Proposition 5.5 (localisation). *For every $(B, W, t) \in S$ we have $B \subseteq \mathcal{R}$, where $|\mathcal{R}| = 10$. Only 153 of the 330 subsets of $\mathcal{R}$ of size 3 or 4 occur as such a B — 83 of size 4 and 70 of size 3 — namely 79 up to the mirror μ . Moreover S is $\hat{\mu}$ -invariant. No condition is imposed on W : each of the 153 shapes occurs in S with between 107 and 1,070 different White configurations.*

Remark 5.6. Black’s men never form the central block B^* of Proposition 8.1 in this particular certificate: the minimisation deleted that shape from the whitelist. This is a fact about the certificate’s local structure only. It does not establish whether either player can force occupation of the block, or force the other player away from it, from the initial position; those reachability claims require a separate proof. The region $\mathcal{R}$ is the neighbourhood of Black’s *half* of the centre, not the centre itself.

Proposition 5.7 (one exchange, and only one). *Let $(B, W, w) \in S$.*

- (i) *If $|B| = 3$ then $\mathcal{Z}((B, W, w), m) = \emptyset$ for every $m \in \mathcal{M}$: White has no capturing move at all. This holds for all 8,968 such positions in S .*
- (ii) *If $|B| = 4$, then of the 18,477 such positions in S exactly 1,999 admit a capturing move for White, and in every one of them every White capture leads to a position of S with $|B| = 3$ (in particular no White capture is winning).*

Corollary 5.8. *In any play from s_0^b in which Black follows a strategy witnessing (S1) for S , White captures at most one Black man in the entire game, and Black is never reduced below three men. Symmetrically for White.*

Proof. By the proof of Theorem 3.2 every position of the play lies in S . Black’s men can only decrease, and by Proposition 5.7(i) once $|B| = 3$ no White move captures ever again; by (ii) the transition from $|B| = 4$ to $|B| = 3$ is not a win for White. $\square$

Thus the certificate encodes a two-layer defence. On the upper layer ($|B| = 4$) Black tolerates White’s capturing chances but only ever offers *bait*: every capture White can make transports the game into the lower layer, where — and this is the substantive content — White’s captures have been extinguished for good. Black’s own winning chances are not renounced: of the positions reachable when Black follows the explicit rule book of Section 7, 6,268 are decisions in which Black’s book move wins outright.

6 Verification

Theorem 5.1 is proved by the program `certify.c` (232 lines of C, no dependencies). It reads the certificate file and performs a single pass over all 3,723,720 positions, checking four conditions. Nothing is iterated and no game values are consulted.

- (C1) *Legality.* Every s with $s \in S$ satisfies $B \cap W = \emptyset$ and $|B|, |W| \in \{3, 4\}$, i.e. $S \subseteq \mathcal{S}$.
- (C2) *Black-safety.* For every Black-to-move $s \in S$, some action is winning or stays in S ; for every White-to-move $s \in S$, no action is winning and all actions stay in S . (Definition 3.1 with $P = \text{Black}$.)
- (C3) *Initial positions.* $s_0^b \in S$ and $s_0^w \in S$.
- (C4) *White-safety of the transferred set.* Independently of Corollary 4.2, the program forms $T = \{s_0^b\} \cup \rho(S)$ — membership in T is decided on the fly by $s \in T \iff \rho s \in S$ or $s = s_0^b$, since $\rho^2 = \text{id}$ — and checks Definition 3.1 for T with $P = \text{White}$.

Check (C4) is logically redundant given (C2), (C3) and Corollary 4.2, whose proof is a page of routine verification; we include it so that the symmetry argument is also machine-checked. The output on `safe6.bin` is

$ S $	91,330
Black shapes occurring in S	153
$ T = \{s_0^b\} \cup \rho(S) $	91,330
(C1) illegal states in S	0
(C2) Black-safety violations	0
(C3) $s_0^b \in S$ and $s_0^w \in S$	yes
(C4) White-safety violations for T	0
running time	0.12 s

That $|T| = |S|$ reflects $\hat{\mu}$ - and ρ -compatibility of S near the initial position: s_0^b already lies in $\rho(S)$ because $s_0^w \in S$.

The action generator inside `certify.c` implements Definition 2.3 directly from the rules text. The Python reference engine is covered by the published unit tests in `test_engine.py`. In particular, the verifier always enumerates both capture candidates, so that (S2) quantifies over White’s choice in the two-candidate case (§8.3) rather than fixing it.

Remark 6.1 (what this verification does, and does not, establish). `certify.c` checks the local certificate conditions, while `test_engine.py` tests the published Python reference implementation and `verify_dual.py` exhaustively checks the rule book in its two colour-conjugate uses. This is not claimed to be cross-validation between two independent retrograde solvers: the only retrograde program in the artifact is the auxiliary, history-free `solve.c`. The remaining risk is a shared misunderstanding of the natural-language rules or a defect in the verification programs. We have not obtained reproduction by an independent third party, nor a specification of Definitions 2.3–3.1 in a proof assistant (Coq, Lean, Isabelle); either would reduce that residual risk.

7 A rule book that works for both colours

A certificate proves that a non-losing strategy exists and, via (S1), implicitly encodes one; but “look up 91,330 bits” is not a strategy a person can follow. We now give an explicit finite object.

7.1 The language

Let $\mathcal{K}$ be the following set of atomic predicates, each a function of the opponent’s set W (and, for the last two, of B as well), for a Black-to-move position (B, W, b) :

$\text{on}(u), \text{off}(u)$	the opponent does / does not occupy square u ;
$\text{count}(k)$	$ W = k$;
$\text{row}_{\geq}(r, k), \text{col}_{\geq}(c, k)$	line r (resp. c) contains at least k opponent men;
$\text{row}_0(r), \text{col}_0(c)$	line r (resp. c) contains no opponent man;
thr, nthr	the opponent does / does not have a capturing move.

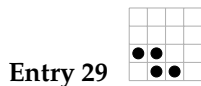
A *rule book* is a finite partial map Π from shapes $B \subseteq X$ to finite sequences of *items*, each item being either a *candidate* $\langle m \rangle$ with m a plain move, or a *conditional* $\langle C \Rightarrow m \rangle$ with $C \subseteq \mathcal{K}$ finite.

The book is executed as follows at a Black-to-move position $s = (B, W, b)$. First $\hat{\mu}$ -canonicalise: if $(\mu B, \mu W)$ precedes (B, W) lexicographically, replace s by $\hat{\mu}s$ and remember to mirror the answer back. Then look up the item sequence of B and scan it in order:

- a candidate $\langle m \rangle$ *fires* if $m \in \mathcal{M}(s)$ and either the resulting action wins or the resulting Black shape is one of the 153 shapes of S ;
- a conditional $\langle C \Rightarrow m \rangle$ *fires* if every predicate in C holds at s (it is guaranteed by construction that then $m \in \mathcal{M}(s)$).

The move of the first item that fires is played, mirrored back if necessary. Finally, if the chosen move has two capture candidates, the one on the larger row index is taken, ties broken towards the smaller column index; this fixes the choice left open by §8.3.

Example 7.1. Entry 29 of Appendix B, for the shape 31, 32, 42, 43, has 7 items:



1. if row 1 has ≥ 2 opp and row 4 empty of opp: 31 $\rightarrow$ 41
2. if opp on 23 and col 4 has ≥ 1 opp: 32 $\rightarrow$ 33
3. if opp on 21 and opp not on 41: 42 $\rightarrow$ 41
4. if col 3 empty of opp and row 1 has ≥ 1 opp: 43 $\rightarrow$ 33
5. 31 $\rightarrow$ 41 (cand.)
6. if opp not on 11 and opp not on 14: 32 $\rightarrow$ 33
7. 43 $\rightarrow$ 44 (cand.)

Read top to bottom. Items 1–4 and 6 are exceptions triggered by specific White placements; items 5 and 7 are the default plan — retreat $31 \rightarrow 41$ if that is legal and keeps a recognised shape, otherwise shuffle $43 \rightarrow 44$ on the bottom rank. The exceptions are listed before the default they override, which is why a rule as unconditional-looking as item 5 can appear in the middle of the list.

The size of the book we obtained is

79 entries, 121 candidates, 849 conditionals, 970 rules in total.

It is printed in full in Appendix B. The machine-readable book is the supplementary file `book_dual.pkl`, executed by the 130-line interpreter `dualbook.py`.

7.2 Extraction

Again the construction is irrelevant to the proof but we record it. We first compute, by policy iteration inside S , for every Black-to-move $s \in S$ the set $G(s)$ of *good* actions — those that win immediately or stay in S — and then look for a small book whose behaviour is pointwise in G . This is a set-cover problem: a candidate item $\langle m \rangle$ placed early in a shape’s list covers all positions of that shape at which m is legal, shape-preserving and good; the remaining positions must be covered by conditionals, which are grown greedily by adding the predicate of $\mathcal{K}$ that covers the most residual positions without covering a position where the move is not good. Iterating — fit the book, recompute the set reachable when the book is followed, refit — converged after a handful of rounds to a book that needs no fallback, i.e. that decides every position reachable under its own play.

7.3 Correctness for both colours

Let Π be the book. For a White-to-move position s define $\Pi^p(s) = \rho_s^{-1}(\Pi(\rho s))$; concretely, rotate the board by 180° , exchange the colours, consult Π , and rotate the answer back. (The capture preference is conjugated accordingly.)

Theorem 7.2. (i) *Following Π from either s_0^b or s_0^w , Black never loses: at every reachable Black-to-move position the book returns a legal move, and at every reachable White-to-move position no White action is winning. The set of positions reachable in this way has 30,245 elements, of which 21,267 are Black decisions and 8,978 are White-to-move; in 6,268 of the decisions Black’s move wins outright by reducing White to two men; White is immobile in 0 of the reachable positions.*

(ii) *Following Π^p from s_0^b , White never loses, in the same sense. The reachable set has 25,274 elements, 17,756 White decisions and 7,518 Black-to-move positions, with 5,912 immediate White wins; Black is immobile in 0 of them.*

(iii) *The reachable set of (ii) is contained in the ρ -image of the reachable set of (i).*

The verification (program `verify_dual.py`) is a breadth-first expansion in which the book’s side plays its single move and the opponent’s side branches over *all* actions, with assertions for legality, for the absence of a winning opposing action, and for non-immobility. It is independent of `certify.c` and of the certificate file: it consults only the book and the rules engine. As an additional cross-check, each successor of a book move was looked up in the value table of Section 8 and never had value “win for the opponent to move” (0 discrepancies out of $21,267 + 17,756$).

The reachable set Σ of Theorem 7.2(i) is by construction Black-safe: Black’s book move witnesses (S1), and the expansion over all White actions witnesses (S2). Since $s_0^b, s_0^w \in \Sigma$ we obtain a second, sharper certificate, which we have verified with the same program `certify.c`:

Theorem 7.3. *The set Σ of positions reachable when Black follows Π from $\{s_0^b, s_0^w\}$ has $|\Sigma| = 30,245$ elements, involves 151 Black shapes, and passes checks (C1)–(C4). Hence Σ alone proves Theorem 5.2, and the strategy witnessing it is the 970-rule book of Appendix B.*

So the certificate can be taken to be Σ , which is 67% smaller than S and comes with a human-executable witness; we retain S because its shape catalogue is the object from which the book was built, and because S is $\hat{\mu}$ -invariant, which Σ need not be.

7.4 The canonical drawn game

If both players follow the book — Black using Π , White using Π^p — the game is decided quickly.

Proposition 7.4. *When Black plays Π and White plays Π^p from s_0^b , the play is*

1.B 42–32 2.W 14–24 3.B 43–42 4.W 12–22 5.B 44–43 6.W 11–12 7.B 43–44 8.W 12–11 9.B 44–43
10.W 11–12 11.B 43–44 12.W 12–11

and is a draw by threefold repetition after 12 plies: the positions reached after plies 4, 8, 12 coincide.

No capture occurs: both sides advance a man to the third rank, bring a second man to the centre, and then shuffle. This is the shape of optimal play in this game — neither side can force an exchange on favourable terms, so both consolidate around the centre and wait. It also illustrates why the repetition rule, and not a capture-based termination condition, is what ends the game.

Remark 7.5. All four of Black’s opening moves preserve the draw, as do (by Lemma 4.1) all four of White’s; so the initial position is not a delicate one. The delicacy is in the middle game: of the 21,267 Black decisions in Σ , 8,974 (42%) admit at least one legal move that loses on the spot, and only 84% of all the legal actions available at these decisions are non-losing. This is why the book needs 849 conditionals: a plan that ignores White’s exact placement will not survive.

8 Corroboration: an auxiliary history-free retrograde classification

Although no part of Theorem 5.2 depends on it, we computed an auxiliary history-free retrograde classification as a sanity check on the certificate and to obtain the local structural facts below. The program `solve.c` performs value iteration over all 3,723,720 positions, assigning *draw* to cycles, and reaches a fixed point in 65 sweeps and 13.7 s. This classification is used only as corroboration and is not claimed to solve the history-sensitive game exactly or to give its value under (T3).

value for the side to move	positions	share
win	2,097,414	56.3%
loss	1,053,186	28.3%
draw	573,120	15.4%
total	3,723,720	

Both s_0^b and s_0^w receive the label *draw*, in agreement with the certificate theorem. Note how sharp this auxiliary classification is: more than five out of six positions receive decisive labels, which is why a drawing strategy must be as detailed as the book of Appendix B.

The table also explains *why* the game is drawn.

Proposition 8.1 (the fortress). *Among the 1,820 four-man shapes $B \subseteq X$, exactly 1 has the property that for every admissible opposing set W Black does not lose, whether Black or White is to move; namely the central block $B^* = \{22, 23, 32, 33\}$. Its profile against the 495 four-man White sets disjoint from it is: Black to move, 489 wins, 6 draws, 0 losses; White to move, 0 wins for White, 40 draws, 455 losses for White. Against the 220 three-man White sets, Black to move wins in all 220, and White to move loses in all 220.*

In words: this is a local structural fact about positions in which a player already holds the central 2×2 block with four men. By the symmetry of Lemma 4.1 the corresponding fact holds for White. It does not imply that either player can force occupation of that block, or force the other player away from it, from the initial position; such reachability claims would require a separate proof. The certificate’s region $\mathcal{R}$ (Proposition 5.5) is a neighbourhood of Black’s half of the centre.

9 Reproducibility

The source code, certificate files, rule book, verification scripts, and expected outputs are released in a public artifact. The artifact is self-contained apart from a standard C compiler and Python 3; the reference run used GCC 11.4 and CPython 3.10 on x86-64 Linux. The fixed reference version is available at commit 30412fb.

To reproduce the complete computational verification of Theorem 5.2, run from the root of that fixed artifact

```
bash scripts/verify.sh
```

This checks every published SHA-256 digest, builds the C verifier, verifies both safety certificates, runs the reference-rule tests, and exhaustively checks the dual rule book. The expected final lines are ALL CERTIFICATE CHECKS PASSED, ALL CHECKS PASSED, and PROOF ARTIFACT VERIFICATION PASSED. For the focused check of the main certificate alone, the corresponding command is

```
cc -O2 -o certify src/certify.c && ./certify artifacts/safe6.bin.
```

Its output has zero violations for (C1), (C2), and (C4), and reports (C3) as 1. The auxiliary retrograde computation is provided only as a consistency check and is not needed for the main proof.

For integrity checking, the published file `safe6.bin` has size 11 328 800 bytes and SHA-256 digest 6DA3BF4F72B850F0167DA092D26E180CDDA2B1D381EA0122F48850985D3B5581. The artifact also includes the exact commands, expected outputs, and checksum file used for the reference run.

10 Discussion

The proof above has three ingredients of different natures, and it is worth separating them. The *mathematical* content is Lemma 2.7 (the repetition rule makes the game finite), Theorem 3.2 (the safety certificate establishes that the constructed positional strategies are non-losing), and Lemma 4.1 together with Corollary 4.2 and Lemma 4.3 (the colour symmetry transfers a certificate from one player to the other, and the initial position is a fixed point of the transfer). The *computational* content is a single pass of local checks over 3,723,720 positions. The *heuristic* content — the search for a small, structured, mirror-symmetric certificate and for a compact rule book — is where all the work went, and none of it needs to be trusted.

Several questions suggest themselves. Is there a certificate small enough to check by hand? Proposition 5.5 reduces Black’s side of the state space to 79 shapes, but each shape still carries thousands of White configurations, and Proposition 5.7 shows that the interaction with White’s configuration cannot be ignored. Our rule book needs 849 conditionals, and we do not know the minimum; the corresponding decision problem (is there a book of $\leq k$ rules in a fixed predicate language?) is a natural measure of a game’s strategic complexity, and is plausibly hard. A second question is scaling: how does the value of the game change on $n \times n$ boards, or with k men each, or if captures are allowed to chain? The mandatory-capture rule makes material a liability as well as an asset — one is sometimes forced to capture into a fork — and it would be interesting to know whether the draw persists.

Author Contributions

The author formulated the game problem and its rules, determined the claims to be established, and chose the certificate-based proof approach. The author developed the mathematical definitions and arguments, proposed and evaluated candidate piece-shape restrictions, interpreted the computational results, and prepared and approved the final manuscript.

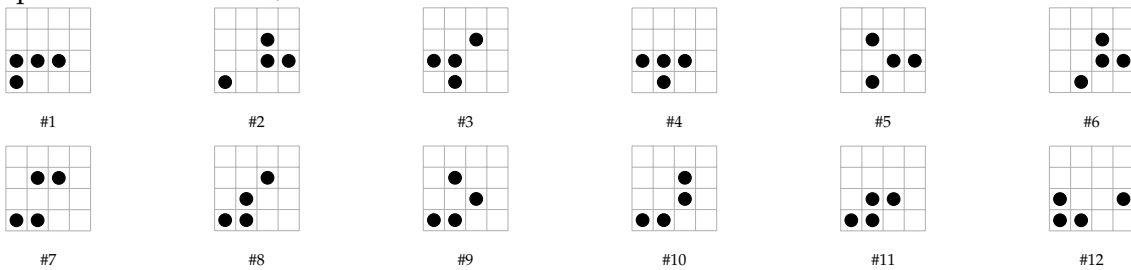
Generative AI Assistance

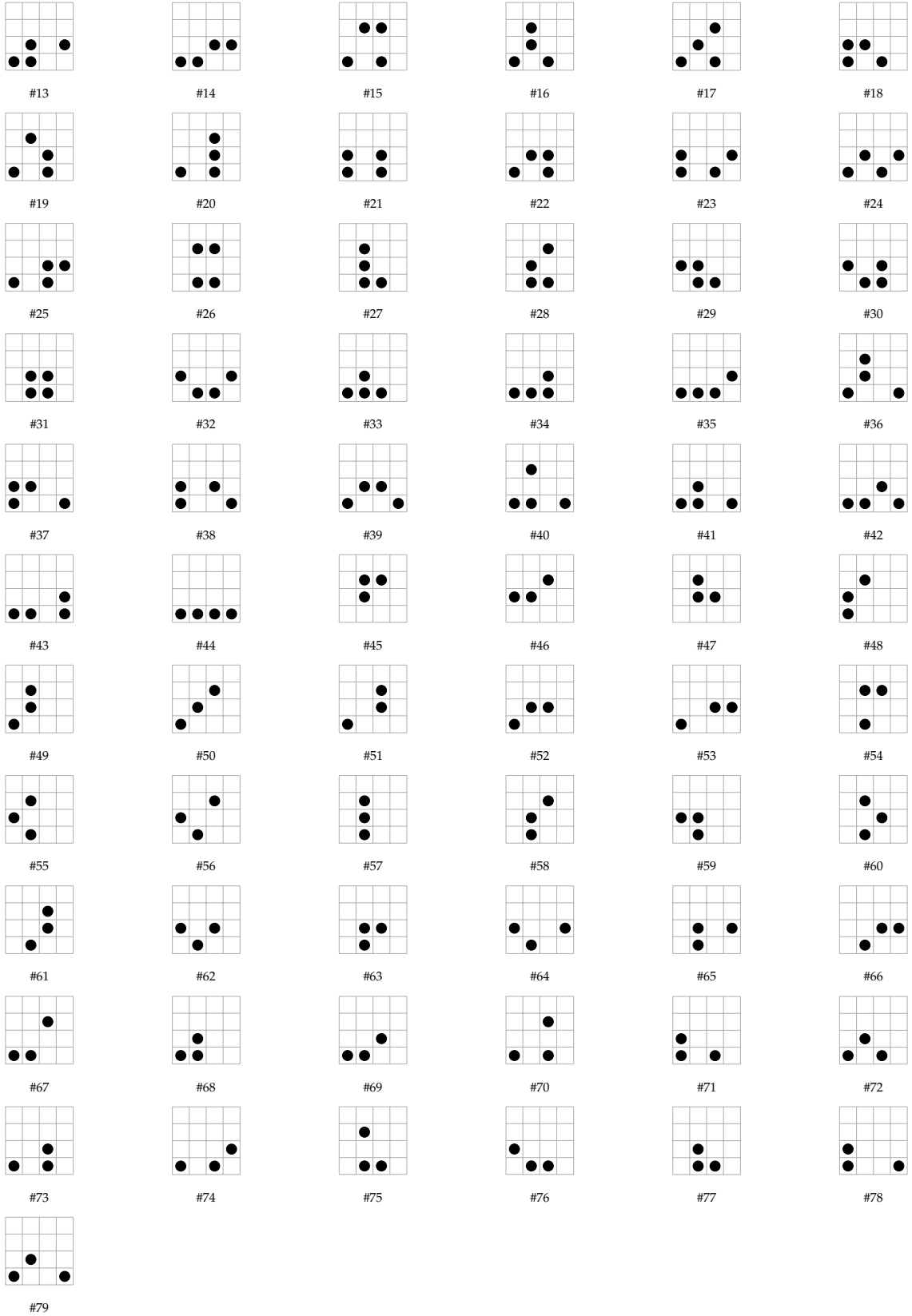
Generative-AI tools were used as research and programming assistants. They contributed to the design of the computational workflow and programs, and were used to generate, modify, debug, and review portions of the C and Python code, as well as to assist with drafting and editing the manuscript. AI-generated suggestions and code were treated as untrusted draft material. The author selected the methods and proof claims, checked the mathematical arguments and implementation, reran and inspected the certificate and rule-book verifications, and takes full responsibility for the final manuscript, the software, and the reported results. The large-scale computations reported here were performed by the deterministic programs released in the public artifact; no AI system is an author.

The two appendices below are data, not argument: nothing in the proof of Theorem 5.2 depends on them (that proof needs only Theorem 5.1 and Section 6), and nothing in Theorem 7.2 or 7.3 depends on the reader inspecting them either, since both are machine-checked in full. They are included so that the certificate and the book are auditable by eye and not only by program; a venue with strict length limits could equally well relegate both to the supplementary material and keep only the worked Example of Section 7 in the printed text, which is the only illustrative material actually used in the argument.

A The shape catalogue

The 79 Black shapes occurring in the certificate S , up to the left–right mirror μ (Proposition 5.5); shapes of four men first, then of three.

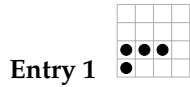




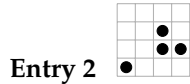
B The rule book

The complete book II: for each of the 79 shapes, the ordered list of items, to be scanned top to bottom. Coordinates are *rc* as in Section 2; “*cand.*” marks a candidate item, which

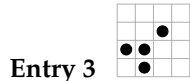
fires when the move is legal and keeps a shape of the catalogue (or wins); all other items are conditionals on the opponent's placement. Black applies the book after mirroring the position into canonical form; White applies it after the colour reversal ρ of Section 4.



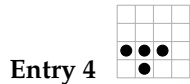
1. if opp not on 13: 41 → 42
2. if opp on 13 and row 1 has ≥ 2 opp: 33 → 43
3. 41 → 42 (cand.)



1. 41 → 42 (cand.)



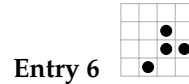
1. if opp on 43 and opp not on 41: 31 → 41
2. if opp not on 13 and col 1 empty of opp: 42 → 43
3. if opp on 14 and opp not on 41: 31 → 41
4. if opp on 34 and row 4 empty of opp: 31 → 41
5. if opp on 21 and col 3 empty of opp: 23 → 33
6. if opp on 24 and opp not on 22: 42 → 43
7. if opp not on 41 and opp has no capturing move: 31 → 41
8. if opp not on 33 and opp not on 44: 23 → 33
9. if opp not on 12 and opp not on 22: 42 → 43
10. 31 → 41 (cand.)
11. 42 → 43 (cand.)



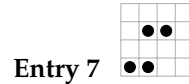
1. if opp not on 41 and col 3 has ≥ 2 opp: 31 → 41
2. if opp on 44 and row 1 has ≥ 2 opp: 33 → 43
3. if opp not on 22 and col 1 empty of opp: 31 → 41
4. if opp not on 43 and col 1 has ≥ 2 opp: 33 → 43
5. if opp on 11 and row 1 has ≥ 2 opp: 31 → 41
6. if opp on 21 and opp on 43: 31 → 41
7. if opp on 21 and opp on 44: 33 → 43
8. if opp on 21 and opp not on 12: 42 → 41
9. if opp not on 23 and col 2 empty of opp: 33 → 23
10. if col 3 empty of opp and row 4 has ≥ 1 opp: 33 → 43
11. if row 4 empty of opp and opp has no capturing move: 31 → 41
12. if opp not on 43 and col 1 empty of opp: 33 → 43
13. if opp on 21 and opp not on 14: 42 → 41
14. if opp not on 41 and opp not on 44: 31 → 41
15. if opp not on 24 and opp not on 43: 33 → 43
16. 33 → 23 (cand.)
17. 31 → 41 (cand.)
18. 42 → 43 (cand.)



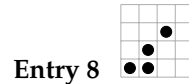
1. if opp on 31 and opp not on 44: 34 → 44
2. if opp not on 13 and opp not on 43: 42 → 43
3. 22 → 23 (cand.)
4. 34 → 44 (cand.)



1. if opp not on 32 and opp not on 44: 34 → 44
2. if opp not on 13 and opp not on 22: 23 → 22
3. if opp not on 43 and row 4 has ≥ 1 opp: 33 → 43
4. 34 → 44 (cand.)



1. if opp not on 11 and opp not on 43: 42 → 43
2. if opp not on 24 and opp not on 33: 23 → 33
3. if opp not on 13 and opp not on 33: 23 → 33
4. if opp on 11 and opp not on 21: 42 → 43
5. 22 → 32 (cand.)



1. if opp on 43 and opp not on 13: 32 → 33
2. if opp on 13 and col 4 has ≥ 2 opp: 32 → 33
3. if row 2 empty of opp and row 4 empty of opp: 42 → 43
4. if col 1 empty of opp and col 3 empty of opp: 42 → 43
5. if row 1 empty of opp and opp has no capturing move: 42 → 43
6. if opp not on 12 and col 1 empty of opp: 41 → 31
7. if col 2 has ≥ 2 opp and opp has no capturing move: 42 → 43
8. if opp on 34 and row 1 has ≥ 2 opp: 23 → 33
9. if opp on 21 and col 4 empty of opp: 32 → 22
10. if opp on 33 and opp has no capturing move: 32 → 22
11. if opp on 33 and opp not on 31: 41 → 31
12. if opp on 34 and opp not on 31: 23 → 33
13. if col 2 empty of opp and row 4 empty of opp: 32 → 22
14. if col 4 has ≥ 2 opp and row 1 has ≥ 2 opp: 23 → 33
15. if opp not on 13 and col 2 empty of opp: 42 → 43
16. if opp not on 14 and row 2 empty of opp: 23 → 33
17. if opp on 12 and opp has a capturing move: 32 → 22

18. if opp on 11 and opp not on 13: 23 → 33
19. if opp not on 13 and opp not on 44: 42 → 43
20. if opp not on 31 and row 4 empty of opp: 32 → 33
21. if opp on 31: 23 → 33
22. if col 2 empty of opp and row 2 has ≥ 1 opp: 32 → 22
23. if opp on 13 and opp on 14: 41 → 31
24. if opp on 11: 32 → 33
25. if opp on 12 and opp on 13: 32 → 22
26. 41 → 31 (cand.)



Entry 9

1. if opp on 23 and opp not on 43: 42 → 43
2. if col 3 has ≥ 1 opp and col 4 has ≥ 2 opp: 22 → 23
3. if opp on 24 and opp not on 21: 33 → 23
4. if opp not on 43 and row 3 empty of opp: 42 → 43
5. if opp on 13 and row 4 empty of opp: 22 → 23
6. if opp not on 21 and row 4 empty of opp: 42 → 43
7. if opp on 34 and opp on 44: 42 → 43
8. if col 2 empty of opp and row 3 has ≥ 1 opp: 22 → 32
9. if opp not on 21 and row 3 empty of opp: 22 → 32
10. if col 3 empty of opp and opp has no capturing move: 42 → 43
11. if opp on 21 and opp on 32: 33 → 23
12. if opp on 32 and opp not on 12: 22 → 23
13. 42 → 43 (cand.)
14. if opp not on 13 and opp not on 23: 22 → 23
15. if opp on 13 and opp on 21: 33 → 23
16. 22 → 32 (cand.)



Entry 10

1. if row 3 has ≥ 1 opp and row 2 empty of opp: 23 → 22
2. if opp on 14 and row 3 empty of opp: 33 → 32
3. if opp on 22 and opp not on 44: 33 → 32
4. if opp not on 13 and row 4 empty of opp: 23 → 22
5. if opp on 11 and opp not on 12: 33 → 32
6. if opp on 44 and opp not on 43: 42 → 43
7. if opp on 43: 23 → 22
8. if opp not on 32 and opp has no capturing move: 33 → 32
9. if opp on 12: 23 → 22
10. 42 → 43 (cand.)



Entry 11

1. if opp on 23 and opp not on 43: 42 → 43
2. if opp on 43 and row 3 empty of opp: 41 → 31
3. if col 2 empty of opp and col 3 empty of opp:

32 → 22

4. if col 1 empty of opp and row 4 empty of opp: 33 → 23
5. if opp not on 43 and col 1 empty of opp: 42 → 43
6. if col 2 empty of opp and row 1 has ≥ 2 opp: 33 → 23
7. if opp on 43 and opp not on 23: 32 → 22
8. if opp on 31 and opp not on 34: 33 → 34
9. if opp not on 21 and row 2 empty of opp: 42 → 43
10. if opp on 22 and row 1 empty of opp: 42 → 43
11. if col 3 has ≥ 1 opp and opp has no capturing move: 32 → 22
12. if opp on 12 and col 4 empty of opp: 42 → 43
13. if opp on 24 and opp on 34: 33 → 23
14. if opp not on 11 and opp not on 12: 42 → 43
15. if opp on 44 and opp not on 22: 32 → 22
16. if opp on 24 and col 4 has ≥ 2 opp: 33 → 23
17. if opp on 22 and opp on 34: 33 → 43
18. if opp not on 13 and opp not on 21: 42 → 43
19. if opp on 22 and opp not on 21: 33 → 43
20. if opp on 11 and opp on 22: 33 → 23
21. if opp on 21 and opp on 24: 33 → 43
22. if opp not on 11 and opp not on 22: 41 → 31
23. if opp on 11 and opp on 21: 32 → 22
24. if opp on 12 and opp not on 44: 42 → 43
25. 41 → 31 (cand.)



Entry 12

1. if opp not on 31 and row 3 empty of opp: 31 → 32
2. if opp on 33 and opp not on 44: 31 → 32
3. 42 → 43 (cand.)



Entry 13

1. if opp on 31 and opp not on 33: 34 → 33
2. if opp on 21 and opp not on 11: 32 → 31
3. if opp not on 21 and col 4 empty of opp: 34 → 44
4. if opp not on 21 and col 3 empty of opp: 42 → 43
5. if opp on 23 and opp not on 43: 42 → 43
6. if opp on 43 and opp not on 44: 34 → 44
7. if opp on 11 and col 2 empty of opp: 42 → 43
8. if opp on 11 and opp not on 44: 34 → 33
9. if opp not on 33 and opp not on 43: 42 → 43
10. 34 → 33 (cand.)
11. if opp not on 13 and row 2 empty of opp: 42 → 43
12. if opp not on 22 and opp not on 31: 32 → 31
13. 34 → 44 (cand.)



Entry 14

1. if opp not on 44 and opp has no capturing

- move: 34 → 44
- if opp not on 24 and opp has no capturing move: 33 → 32
 - if col 1 empty of opp and row 4 empty of opp: 42 → 43
 - if col 3 empty of opp and row 2 has ≥ 1 opp: 33 → 43
 - if opp not on 12 and col 3 has ≥ 2 opp: 34 → 44
 - if opp not on 22 and col 3 has ≥ 1 opp: 42 → 43
 - if opp not on 23 and col 1 has ≥ 1 opp: 33 → 32
 - 33 → 43 (cand.)



Entry 15

- if opp on 44 and opp not on 42: 41 → 42
- if opp not on 22 and row 3 empty of opp: 23 → 33
- if opp on 34 and col 3 empty of opp: 23 → 33
- if opp not on 32 and row 2 empty of opp: 22 → 32
- if opp not on 24 and row 1 has ≥ 1 opp: 41 → 42
- if opp not on 11 and col 2 empty of opp: 41 → 42
- if opp on 32 and row 2 has ≥ 1 opp: 41 → 42
- if opp on 24 and opp not on 33: 22 → 32
- if opp not on 21 and opp not on 34: 43 → 42
- 22 → 32 (cand.)



Entry 16

- if row 3 empty of opp and opp has no capturing move: 32 → 33
- if row 2 empty of opp and opp has no capturing move: 22 → 23
- if opp on 23 and opp not on 33: 32 → 33
- if opp on 24 and opp not on 33: 22 → 23
- if opp not on 31 and opp not on 34: 41 → 31
- if opp on 34 and opp not on 33: 32 → 33
- 32 → 31 (cand.)
- if opp not on 23 and opp not on 44: 22 → 23
- 41 → 42 (cand.)



Entry 17

- if col 2 empty of opp and row 3 empty of opp: 23 → 33
- if row 2 empty of opp and row 3 empty of opp: 23 → 22
- if opp on 44 and row 1 has ≥ 2 opp: 41 → 42
- if col 1 empty of opp and col 2 empty of opp: 32 → 22
- if col 3 empty of opp and col 4 empty of opp: 23 → 33
- if opp on 34 and col 2 has ≥ 2 opp: 23 → 33

- if col 3 empty of opp and row 2 empty of opp: 23 → 22
- if opp on 31 and opp on 44: 41 → 42
- if opp on 34 and row 2 has ≥ 2 opp: 43 → 42
- if opp on 21 and opp on 34: 32 → 22
- if opp on 24 and opp on 34: 32 → 22
- if row 1 has ≥ 2 opp and row 2 has ≥ 2 opp: 41 → 42
- if opp on 24 and row 1 has ≥ 2 opp: 32 → 22
- if opp on 34 and opp has a capturing move: 41 → 42
- if col 3 empty of opp and row 3 has ≥ 1 opp: 23 → 33
- if opp on 33 and col 2 has ≥ 2 opp: 41 → 31
- if opp on 21 and row 1 has ≥ 2 opp: 32 → 22
- if opp not on 22 and row 3 empty of opp: 23 → 22
- if opp not on 31 and row 1 empty of opp: 41 → 31
- if opp not on 33 and col 4 empty of opp: 23 → 33
- if opp on 22 and opp not on 14: 41 → 42
- if opp on 21 and opp not on 31: 41 → 31
- if opp not on 14 and opp not on 33: 23 → 22
- if col 1 empty of opp and row 4 has ≥ 1 opp: 41 → 31
- if opp not on 24 and col 1 empty of opp: 43 → 42
- if opp not on 22 and row 2 has ≥ 1 opp: 32 → 22
- if opp not on 44 and col 2 empty of opp: 43 → 42
- if opp on 31 and opp not on 22: 23 → 22
- if opp not on 22 and opp not on 34: 41 → 31
- 32 → 22 (cand.)
- if opp not on 12 and opp not on 24: 41 → 42
- 23 → 33 (cand.)
- 41 → 31 (cand.)



Entry 18

- if opp not on 23: 41 → 42
- 32 → 33 (cand.)



Entry 19

- if opp on 44 and col 2 empty of opp: 41 → 42
- if opp on 44 and row 3 empty of opp: 33 → 32
- if opp on 44 and opp not on 32: 22 → 32
- if opp on 44 and opp has no capturing move: 41 → 42
- if opp on 42 and col 4 empty of opp: 22 → 32
- if opp not on 23 and row 1 empty of opp: 33 → 23
- if opp on 34 and row 2 has ≥ 2 opp: 22 → 32
- if opp on 24 and opp on 34: 33 → 23
- if col 2 empty of opp and col 3 empty of opp: 41 → 42

10. if opp on 12 and row 2 has ≥ 2 opp: 33 $\rightarrow$ 32
11. if opp on 12 and opp on 21: 33 $\rightarrow$ 23
12. if opp on 12 and opp on 24: 33 $\rightarrow$ 23
13. if col 2 has ≥ 1 opp and row 3 empty of opp: 33 $\rightarrow$ 32
14. if row 3 empty of opp and opp has no capturing move: 41 $\rightarrow$ 42
15. if opp not on 32 and row 2 has ≥ 1 opp: 22 $\rightarrow$ 32
16. if opp on 14 and opp not on 32: 22 $\rightarrow$ 32
17. if opp not on 34 and row 2 empty of opp: 41 $\rightarrow$ 42
18. if opp not on 23 and row 2 has ≥ 1 opp: 33 $\rightarrow$ 23
19. if opp not on 32 and opp has no capturing move: 33 $\rightarrow$ 32
20. if opp not on 42 and col 4 has ≥ 1 opp: 43 $\rightarrow$ 42
21. 33 $\rightarrow$ 32 (cand.)
22. if opp not on 11 and opp not on 31: 41 $\rightarrow$ 42
23. 43 $\rightarrow$ 42 (cand.)



Entry 20

1. if opp not on 31 and col 2 empty of opp: 23 $\rightarrow$ 22
2. if opp not on 13 and row 2 empty of opp: 23 $\rightarrow$ 22
3. if opp on 14 and row 3 empty of opp: 33 $\rightarrow$ 32
4. if opp not on 32 and row 1 empty of opp: 33 $\rightarrow$ 32
5. if opp on 44 and opp not on 13: 41 $\rightarrow$ 42
6. if opp on 11 and row 4 empty of opp: 43 $\rightarrow$ 42
7. if opp on 22 and opp not on 12: 41 $\rightarrow$ 42
8. if opp not on 32 and col 1 empty of opp: 33 $\rightarrow$ 32
9. if opp on 12 and opp not on 31: 23 $\rightarrow$ 22
10. if opp on 13 and opp not on 44: 43 $\rightarrow$ 42
11. 33 $\rightarrow$ 32 (cand.)
12. if opp on 32 and opp not on 21: 41 $\rightarrow$ 42
13. 23 $\rightarrow$ 22 (cand.)



Entry 21

1. if opp on 34 and opp not on 32: 31 $\rightarrow$ 32
2. if opp not on 42 and col 4 has ≥ 2 opp: 41 $\rightarrow$ 42
3. if opp not on 12 and col 2 empty of opp: 41 $\rightarrow$ 42
4. if opp not on 32 and opp not on 44: 31 $\rightarrow$ 32
5. 43 $\rightarrow$ 44 (cand.)
6. if opp not on 22 and opp not on 23: 33 $\rightarrow$ 34
7. if opp not on 13 and opp not on 42: 41 $\rightarrow$ 42
8. if opp on 13 and opp not on 12: 41 $\rightarrow$ 42
9. if opp on 22: 31 $\rightarrow$ 32
10. 41 $\rightarrow$ 42 (cand.)
11. 31 $\rightarrow$ 32 (cand.)



Entry 22

1. if opp on 44 and opp not on 42: 41 $\rightarrow$ 42
2. if opp not on 31 and row 2 empty of opp: 32 $\rightarrow$ 22
3. if opp not on 42 and row 1 empty of opp: 43 $\rightarrow$ 42
4. if opp not on 34 and row 2 empty of opp: 33 $\rightarrow$ 34
5. if opp not on 44 and row 1 empty of opp: 43 $\rightarrow$ 44
6. if opp on 42 and col 2 has ≥ 2 opp: 33 $\rightarrow$ 34
7. if opp on 22 and opp not on 12: 41 $\rightarrow$ 42
8. if opp not on 23 and col 1 empty of opp: 33 $\rightarrow$ 23
9. if opp on 34 and row 2 has ≥ 2 opp: 32 $\rightarrow$ 31
10. if col 2 empty of opp and row 2 has ≥ 2 opp: 32 $\rightarrow$ 22
11. if opp on 21 and opp has a capturing move: 32 $\rightarrow$ 22
12. if col 2 empty of opp and row 3 empty of opp: 32 $\rightarrow$ 42
13. if opp not on 31 and col 2 empty of opp: 32 $\rightarrow$ 31
14. if col 1 empty of opp and opp has a capturing move: 32 $\rightarrow$ 42
15. if opp not on 23 and col 4 empty of opp: 33 $\rightarrow$ 34
16. if opp not on 14 and col 2 empty of opp: 41 $\rightarrow$ 42
17. if col 4 empty of opp and row 1 has ≥ 2 opp: 41 $\rightarrow$ 42
18. if opp not on 44 and col 1 empty of opp: 33 $\rightarrow$ 34
19. if opp on 42 and opp not on 11: 32 $\rightarrow$ 31
20. if col 2 has ≥ 1 opp and col 4 has ≥ 2 opp: 33 $\rightarrow$ 23
21. if opp not on 34 and col 3 empty of opp: 33 $\rightarrow$ 34
22. if opp on 23 and opp has no capturing move: 43 $\rightarrow$ 44
23. if opp not on 11 and opp not on 13: 43 $\rightarrow$ 42
24. if opp on 23 and opp not on 24: 32 $\rightarrow$ 42
25. if opp not on 13 and opp not on 22: 32 $\rightarrow$ 22
26. if opp on 11 and opp on 13: 33 $\rightarrow$ 23
27. 33 $\rightarrow$ 34 (cand.)
28. if opp on 12: 43 $\rightarrow$ 44
29. 43 $\rightarrow$ 42 (cand.)



Entry 23

1. if opp on 11 and opp not on 42: 41 $\rightarrow$ 42
2. if opp on 23 and opp not on 13: 34 $\rightarrow$ 33
3. if opp on 44 and opp not on 42: 41 $\rightarrow$ 42
4. if opp not on 21 and row 3 empty of opp: 31 $\rightarrow$ 32
5. if opp on 21 and opp not on 33: 34 $\rightarrow$ 33

6. if opp on 33 and opp has a capturing move:
43 → 42
7. 31 → 32 (cand.)
8. if opp on 32 and opp not on 14: 34 → 33
9. if opp on 14 and opp not on 12: 41 → 42
10. 34 → 33 (cand.)



Entry 24

1. if opp not on 42 and col 3 empty of opp:
41 → 42
2. if opp not on 33 and col 2 empty of opp:
34 → 33
3. if opp on 31 and opp not on 33: 34 → 33
4. if opp on 42 and col 4 empty of opp: 34 → 44
5. if opp not on 44 and col 3 empty of opp:
34 → 44
6. if opp not on 13 and opp not on 33: 34 → 33
7. if opp on 44 and opp not on 13: 41 → 42
8. if opp on 24 and opp on 33: 43 → 44
9. if opp on 33 and opp has no capturing move:
34 → 44
10. if opp on 11 and opp on 22: 41 → 42
11. if opp not on 44 and opp has no capturing move:
34 → 44
12. if opp on 21 and opp not on 11: 32 → 31
13. if opp on 13 and opp has no capturing move:
34 → 33
14. if opp not on 12 and row 2 has ≥2 opp:
41 → 42
15. if opp on 22 and opp not on 12: 43 → 42
16. if opp on 13 and opp on 24: 43 → 44
17. if opp on 13 and opp on 23: 32 → 33
18. if opp on 33 and row 1 has ≥2 opp: 34 → 44
19. if opp on 33 and col 3 has ≥2 opp: 43 → 42
20. if opp on 11: 41 → 42
21. 34 → 44 (cand.)



Entry 25

1. if opp on 44 and opp not on 22: 41 → 42
2. if opp not on 32 and col 3 empty of opp:
33 → 32
3. if opp not on 31 and col 4 empty of opp:
34 → 44
4. if row 3 empty of opp and opp has no capturing move:
33 → 32
5. if opp on 14 and opp not on 11: 34 → 44
6. if opp on 24 and opp not on 11: 43 → 44
7. if opp on 31 and row 1 has ≥1 opp: 41 → 42
8. if opp not on 13 and opp not on 22: 34 → 44
9. if opp on 23: 43 → 42
10. 34 → 44 (cand.)



Entry 26

1. if opp not on 14 and opp not on 32: 22 → 32
2. if opp not on 31 and opp not on 41: 42 → 41

3. if opp on 14 and col 2 empty of opp: 22 → 32
4. if opp on 41 and opp not on 44: 43 → 44
5. 23 → 33 (cand.)
6. 42 → 41 (cand.)



Entry 27

1. if opp not on 12 and row 2 empty of opp:
22 → 23
2. if opp on 23 and col 4 has ≥1 opp: 32 → 33
3. if opp on 24 and opp not on 12: 22 → 23
4. if opp on 14 and row 4 empty of opp: 42 → 41
5. if opp not on 13 and row 3 empty of opp:
32 → 33
6. if opp on 41 and opp not on 24: 43 → 44
7. if opp on 21 and opp not on 33: 32 → 33
8. if opp not on 11 and opp has no capturing move:
42 → 41
9. if opp not on 31 and opp has no capturing move:
42 → 41
10. if opp on 13 and opp not on 23: 22 → 23
11. if opp not on 33 and opp not on 44: 32 → 33
12. 43 → 44 (cand.)
13. 42 → 41 (cand.)



Entry 28

1. if opp not on 22 and opp has a capturing move:
32 → 22
2. if opp not on 22 and col 1 empty of opp:
42 → 41
3. if col 3 empty of opp and col 4 empty of opp:
23 → 33
4. if opp not on 13 and col 1 empty of opp:
42 → 41
5. if row 2 empty of opp and row 3 empty of opp:
32 → 33
6. if opp on 31 and row 2 has ≥2 opp: 23 → 33
7. if opp on 24 and opp on 31: 32 → 22
8. if opp not on 41 and row 1 empty of opp:
42 → 41
9. if row 1 has ≥2 opp and row 2 has ≥2 opp:
43 → 44
10. if opp not on 22 and row 3 empty of opp:
32 → 22
11. if opp on 24 and opp on 34: 23 → 33
12. if opp on 44 and col 1 has ≥2 opp: 23 → 33
13. if opp on 24 and opp not on 44: 43 → 44
14. if opp on 22 and col 1 has ≥2 opp: 23 → 33
15. if opp on 11 and opp on 21: 32 → 22
16. if opp on 11 and opp on 31: 43 → 44
17. if col 2 has ≥2 opp and row 4 empty of opp:
23 → 33
18. if opp on 12 and opp not on 41: 42 → 41
19. if col 3 empty of opp and row 2 empty of opp:
23 → 33
20. if opp on 13 and col 4 empty of opp: 43 → 44
21. if opp not on 22 and col 3 empty of opp:

- 32 → 22
22. if opp not on 22 and col 4 empty of opp: 23 → 22
 23. if opp on 21 and opp not on 22: 32 → 22
 24. if opp on 41 and opp not on 22: 43 → 44
 25. if opp on 34 and row 4 has ≥ 1 opp: 23 → 33
 26. if opp on 33 and col 4 has ≥ 1 opp: 42 → 41
 27. if opp on 34 and opp not on 31: 23 → 33
 28. if opp not on 44 and opp has no capturing move: 43 → 44
 29. if opp not on 41 and col 3 empty of opp: 42 → 41
 30. if opp on 44 and opp not on 14: 32 → 33
 31. if opp on 13 and opp on 31: 23 → 33
 32. 43 → 44 (cand.)
 33. 42 → 41 (cand.)



Entry 29

1. if row 1 has ≥ 2 opp and row 4 empty of opp: 31 → 41
2. if opp on 23 and col 4 has ≥ 1 opp: 32 → 33
3. if opp on 21 and opp not on 41: 42 → 41
4. if col 3 empty of opp and row 1 has ≥ 1 opp: 43 → 33
5. 31 → 41 (cand.)
6. if opp not on 11 and opp not on 14: 32 → 33
7. 43 → 44 (cand.)



Entry 30

1. if opp on 34 and opp not on 32: 31 → 32
2. if opp on 21 and opp has a capturing move: 42 → 41
3. if opp on 14 and opp on 22: 31 → 32
4. if opp not on 32 and row 2 empty of opp: 33 → 34
5. if col 2 empty of opp and col 3 empty of opp: 33 → 34
6. if opp on 22 and opp on 23: 31 → 32
7. if opp not on 44 and col 3 has ≥ 2 opp: 43 → 44
8. if opp on 21 and opp on 22: 31 → 32
9. if opp on 14 and opp on 44: 42 → 41
10. if opp on 41 and col 4 empty of opp: 43 → 44
11. if opp on 13 and opp on 14: 33 → 34
12. if opp on 11 and opp on 14: 31 → 41
13. if col 4 has ≥ 2 opp and col 2 empty of opp: 42 → 41
14. if opp not on 34 and row 1 empty of opp: 33 → 34
15. if opp on 22 and col 4 has ≥ 1 opp: 33 → 32
16. if opp on 44 and opp not on 12: 42 → 41
17. if opp on 14 and col 1 has ≥ 1 opp: 43 → 44
18. if opp not on 32 and col 3 empty of opp: 33 → 34
19. if opp on 32 and col 1 has ≥ 1 opp: 43 → 44
20. if opp not on 22 and col 4 empty of opp:

- 31 → 41
21. if opp on 23 and col 1 has ≥ 1 opp: 33 → 34
 22. if opp on 12 and opp on 23: 31 → 41
 23. if opp not on 11 and opp not on 32: 31 → 32
 24. if opp on 14 and opp not on 24: 42 → 41
 25. if opp not on 13 and opp not on 34: 33 → 34
 26. if opp on 11 and opp on 22: 31 → 32
 27. if opp on 11 and opp not on 41: 31 → 41
 28. 33 → 32 (cand.)
 29. if col 3 has ≥ 1 opp and row 2 has ≥ 1 opp: 42 → 41
 30. if opp on 32 and opp not on 12: 43 → 44
 31. 42 → 41 (cand.)



Entry 31

1. if opp not on 44 and row 2 empty of opp: 43 → 44
2. if opp not on 44 and row 1 empty of opp: 43 → 44
3. if col 3 empty of opp and row 1 has ≥ 2 opp: 43 → 44
4. if opp not on 31 and col 3 empty of opp: 33 → 23
5. if col 3 empty of opp and col 4 empty of opp: 33 → 34
6. if opp on 21 and opp on 24: 33 → 23
7. if opp on 12 and opp on 24: 33 → 34
8. if opp on 24 and opp on 41: 33 → 23
9. if opp on 24 and opp not on 22: 33 → 34
10. if opp on 23 and row 3 has ≥ 1 opp: 43 → 44
11. if opp not on 22 and opp not on 41: 32 → 22
12. if opp on 13 and opp not on 22: 43 → 44
13. if opp on 21 and opp not on 23: 33 → 23
14. if opp on 13 and opp not on 23: 33 → 34
15. if opp not on 12 and col 4 empty of opp: 43 → 44
16. if opp on 31 and opp on 44: 33 → 23
17. if opp on 41 and opp not on 21: 43 → 44
18. 32 → 22 (cand.)
19. if opp on 23 and opp not on 11: 32 → 31
20. if opp on 14: 33 → 23
21. if opp on 11 and opp on 24: 33 → 34
22. 42 → 41 (cand.)



Entry 32

1. if opp on 24 and opp not on 14: 43 → 44
2. if opp on 23 and opp not on 13: 34 → 33
3. if opp on 11 and opp not on 32: 31 → 32
4. if opp on 22 and opp not on 32: 31 → 32
5. if opp on 21 and opp not on 41: 42 → 41
6. if opp not on 32 and col 4 empty of opp: 43 → 44
7. if opp on 21 and opp not on 12: 34 → 33
8. if row 2 empty of opp and row 3 empty of opp: 34 → 33
9. if opp not on 23 and opp not on 32: 43 → 44

10. if opp not on 12 and col 4 has ≥ 1 opp: 34 $\rightarrow$ 33
11. if opp on 11 and opp not on 41: 31 $\rightarrow$ 41
12. if opp on 32 and opp not on 12: 43 $\rightarrow$ 44
13. if opp not on 13 and opp not on 14: 34 $\rightarrow$ 33
14. if opp on 12 and opp on 13: 42 $\rightarrow$ 41
15. 31 $\rightarrow$ 41 (cand.)



Entry 33

1. if col 3 empty of opp and row 3 has ≥ 1 opp: 43 $\rightarrow$ 33
2. if col 2 empty of opp and row 3 empty of opp: 32 $\rightarrow$ 33
3. if opp not on 33 and col 2 empty of opp: 43 $\rightarrow$ 33
4. if opp on 33 and row 1 has ≥ 2 opp: 41 $\rightarrow$ 31
5. if row 1 has ≥ 2 opp and row 3 has ≥ 1 opp: 43 $\rightarrow$ 33
6. if opp on 22 and col 4 empty of opp: 43 $\rightarrow$ 44
7. if opp not on 22 and col 3 empty of opp: 32 $\rightarrow$ 33
8. if opp on 33 and opp on 44: 41 $\rightarrow$ 31
9. if opp on 33 and row 3 has ≥ 2 opp: 43 $\rightarrow$ 44
10. if opp on 33: 41 $\rightarrow$ 31
11. if opp not on 21 and opp not on 22: 32 $\rightarrow$ 33
12. if opp on 23 and col 1 empty of opp: 32 $\rightarrow$ 33
13. if opp not on 13 and col 1 empty of opp: 43 $\rightarrow$ 33
14. if opp on 14 and opp not on 44: 43 $\rightarrow$ 44
15. if opp not on 12 and opp not on 23: 41 $\rightarrow$ 31
16. if opp on 21 and opp not on 13: 32 $\rightarrow$ 33
17. if opp on 11 and col 3 has ≥ 1 opp: 32 $\rightarrow$ 33
18. if opp on 12 and opp not on 44: 43 $\rightarrow$ 44
19. 41 $\rightarrow$ 31 (cand.)
20. 43 $\rightarrow$ 33 (cand.)



Entry 34

1. if opp not on 32 and row 3 has ≥ 1 opp: 42 $\rightarrow$ 32
2. if opp on 44 and col 2 empty of opp: 42 $\rightarrow$ 32
3. if opp not on 12 and col 2 empty of opp: 41 $\rightarrow$ 31
4. if opp not on 12 and opp not on 32: 33 $\rightarrow$ 32
5. if opp not on 32 and row 2 empty of opp: 42 $\rightarrow$ 32
6. if opp on 32 and col 3 has ≥ 1 opp: 43 $\rightarrow$ 44
7. if opp on 22 and col 1 has ≥ 1 opp: 42 $\rightarrow$ 32
8. if opp not on 34 and row 2 empty of opp: 33 $\rightarrow$ 34
9. if opp on 11 and opp not on 24: 41 $\rightarrow$ 31
10. if opp on 23 and opp not on 13: 33 $\rightarrow$ 32
11. if opp on 24 and opp on 44: 33 $\rightarrow$ 34
12. if opp not on 12 and col 1 empty of opp: 41 $\rightarrow$ 31
13. if opp on 32 and opp not on 21: 43 $\rightarrow$ 44
14. if opp on 14 and opp on 21: 33 $\rightarrow$ 34

15. if opp on 22 and opp not on 14: 42 $\rightarrow$ 32
16. if opp on 14 and opp not on 22: 41 $\rightarrow$ 31
17. if opp on 21 and opp on 44: 33 $\rightarrow$ 34
18. if opp on 11 and opp not on 21: 42 $\rightarrow$ 32
19. if opp on 13 and opp not on 23: 33 $\rightarrow$ 32
20. if opp on 12 and opp not on 44: 43 $\rightarrow$ 44
21. if opp on 21 and opp on 24: 33 $\rightarrow$ 34
22. 43 $\rightarrow$ 44 (cand.)
23. if opp on 12 and opp on 14: 42 $\rightarrow$ 32
24. 41 $\rightarrow$ 31 (cand.)



Entry 35

1. if col 2 empty of opp and opp has no capturing move: 42 $\rightarrow$ 32
2. if opp on 32 and opp not on 31: 43 $\rightarrow$ 33
3. if opp on 22 and col 3 empty of opp: 42 $\rightarrow$ 32
4. if opp on 23: 34 $\rightarrow$ 33
5. if opp on 12 and row 2 empty of opp: 42 $\rightarrow$ 32
6. if opp not on 44 and opp has a capturing move: 43 $\rightarrow$ 44
7. if opp not on 21 and opp not on 32: 42 $\rightarrow$ 32
8. if opp on 21 and row 2 has ≥ 2 opp: 41 $\rightarrow$ 31
9. if opp not on 13 and opp not on 24: 34 $\rightarrow$ 33
10. 42 $\rightarrow$ 32 (cand.)
11. 43 $\rightarrow$ 44 (cand.)



Entry 36

1. if opp not on 13 and opp not on 43: 44 $\rightarrow$ 43
2. if opp not on 12 and opp not on 33: 41 $\rightarrow$ 42
3. if opp on 13 and opp not on 21: 41 $\rightarrow$ 42
4. if opp on 13 and opp not on 23: 41 $\rightarrow$ 31
5. 32 $\rightarrow$ 42 (cand.)



Entry 37

1. 41 $\rightarrow$ 42 (cand.)



Entry 38

1. if opp not on 21 and opp not on 32: 31 $\rightarrow$ 32
2. if opp not on 11 and opp not on 42: 44 $\rightarrow$ 43
3. if opp on 21 and opp not on 12: 31 $\rightarrow$ 32
4. 41 $\rightarrow$ 42 (cand.)



Entry 39

1. if col 2 has ≥ 1 opp and opp has no capturing move: 33 $\rightarrow$ 43
2. if opp on 23 and opp not on 13: 44 $\rightarrow$ 43
3. if opp not on 13 and opp not on 42: 41 $\rightarrow$ 42
4. if opp on 42: 33 $\rightarrow$ 43
5. if opp on 22 and opp not on 12: 41 $\rightarrow$ 42
6. if opp not on 24 and opp has no capturing move: 32 $\rightarrow$ 42
7. if opp not on 22 and opp not on 24: 41 $\rightarrow$ 42
8. if opp on 13 and opp on 24: 33 $\rightarrow$ 34

9. if opp on 12 and opp not on 21: 44 → 43
10. 33 → 43 (cand.)



Entry 40

1. if opp on 12: 42 → 32
2. if opp not on 12 and opp not on 24: 42 → 32
3. 22 → 32 (cand.)



Entry 41

1. if opp on 43 and opp not on 33: 32 → 33
2. if col 3 empty of opp and row 3 empty of opp: 32 → 33
3. if opp not on 13 and col 3 empty of opp: 44 → 43
4. if col 2 empty of opp and row 3 empty of opp: 42 → 43
5. if opp on 33 and row 1 has ≥ 2 opp: 41 → 31
6. if opp on 21 and row 3 has ≥ 1 opp: 41 → 31
7. if opp not on 33 and row 2 empty of opp: 32 → 33
8. if opp not on 33 and col 3 has ≥ 2 opp: 32 → 33
9. if opp not on 12 and row 3 empty of opp: 44 → 43
10. if col 1 empty of opp and col 4 empty of opp: 41 → 31
11. if opp on 31 and col 4 has ≥ 1 opp: 42 → 43
12. if opp on 24 and opp on 33: 41 → 31
13. if opp not on 11 and col 2 has ≥ 2 opp: 42 → 43
14. if opp on 21 and opp not on 11: 44 → 43
15. if opp on 33 and opp has a capturing move: 44 → 43
16. if opp not on 43 and col 3 has ≥ 2 opp: 42 → 43
17. if opp on 23 and opp on 34: 32 → 33
18. if opp not on 14 and col 4 has ≥ 1 opp: 44 → 43
19. if opp on 33 and opp not on 31: 41 → 31
20. if opp on 22 and opp on 23: 42 → 43
21. if opp on 23 and opp not on 11: 44 → 34
22. if opp on 12 and opp not on 11: 42 → 43
23. if opp on 12 and opp not on 22: 32 → 22
24. 41 → 31 (cand.)
25. if opp on 31 and opp not on 11: 44 → 34
26. 32 → 33 (cand.)



Entry 42

1. if opp not on 32 and row 3 has ≥ 1 opp: 42 → 32
2. if col 2 empty of opp and opp has no capturing move: 41 → 31
3. if row 1 has ≥ 2 opp and row 2 empty of opp: 44 → 34
4. if opp on 32 and col 3 has ≥ 2 opp: 44 → 34

5. if opp on 23 and opp has a capturing move: 42 → 43
6. if opp on 23 and opp not on 12: 33 → 32
7. if opp on 13 and col 2 empty of opp: 44 → 43
8. if opp not on 12 and opp not on 32: 33 → 32
9. if opp not on 12 and col 4 empty of opp: 44 → 34
10. if opp on 21 and opp on 43: 44 → 34
11. if col 1 has ≥ 1 opp and row 2 empty of opp: 42 → 43
12. if opp on 11 and opp on 32: 44 → 34
13. if opp on 24 and opp on 43: 33 → 34
14. if opp on 32 and row 2 has ≥ 2 opp: 44 → 34
15. if opp on 11 and opp on 21: 44 → 43
16. if opp on 11 and opp not on 13: 42 → 32
17. if opp on 22 and opp has a capturing move: 42 → 43
18. if opp on 13 and opp on 24: 33 → 34
19. if opp not on 34 and opp has a capturing move: 44 → 34
20. if opp on 22 and row 1 has ≥ 2 opp: 42 → 32
21. if col 1 empty of opp and opp has no capturing move: 41 → 31
22. if opp on 21 and opp not on 24: 42 → 43
23. if opp not on 14 and opp not on 32: 41 → 31
24. if opp on 14: 44 → 43
25. 33 → 34 (cand.)
26. 42 → 43 (cand.)



Entry 43

1. if opp not on 23 and row 3 empty of opp: 41 → 31
2. if opp on 33: 42 → 32
3. if opp not on 21 and row 3 has ≥ 1 opp: 34 → 33
4. if opp not on 12 and opp not on 14: 34 → 33
5. if opp on 23 and opp not on 43: 41 → 31
6. 44 → 43 (cand.)
7. 34 → 33 (cand.)



Entry 44

1. if opp not on 23 and row 2 has ≥ 1 opp: 44 → 34
2. 42 → 32 (cand.)
3. 43 → 33 (cand.)



Entry 45

1. if opp not on 24 and col 3 empty of opp: 23 → 33
2. if opp not on 11 and col 3 has ≥ 1 opp: 32 → 33
3. if opp not on 14 and col 3 empty of opp: 23 → 33
4. if opp not on 42 and opp not on 43: 32 → 42
5. if opp on 11 and opp not on 13: 32 → 33

6. 32 → 42 (cand.)
7. 32 → 33 (cand.)



Entry 46

1. if opp not on 12 and opp not on 22: 31 → 41
2. 23 → 22 (cand.)
3. if opp on 22 and opp not on 43: 31 → 41
4. 32 → 33 (cand.)



Entry 47

1. if col 2 empty of opp and row 2 empty of opp: 22 → 23
2. if opp on 24 and row 4 has ≥ 1 opp: 22 → 23
3. if opp not on 23 and col 1 has ≥ 2 opp: 33 → 23
4. if opp not on 42 and col 3 empty of opp: 32 → 42
5. if opp not on 23 and row 2 has ≥ 1 opp: 33 → 23
6. if opp not on 43 and row 1 has ≥ 2 opp: 33 → 43
7. if opp on 41 and opp not on 43: 33 → 43
8. if opp on 42 and col 1 empty of opp: 22 → 23
9. if opp on 11 and opp not on 42: 32 → 42
10. if opp on 43 and col 3 has ≥ 2 opp: 32 → 42
11. if opp not on 23 and col 3 has ≥ 1 opp: 22 → 23
12. if opp not on 13 and row 3 empty of opp: 33 → 43
13. if opp on 23 and opp not on 13: 32 → 42
14. if opp on 42 and opp not on 12: 22 → 23
15. 33 → 23 (cand.)
16. 33 → 43 (cand.)



Entry 48

1. if opp not on 13 and opp not on 32: 31 → 32
2. if opp on 11: 41 → 42
3. 31 → 32 (cand.)



Entry 49

1. if opp on 21 and opp not on 31: 32 → 31
2. if opp not on 21 and opp has no capturing move: 22 → 23
3. if opp not on 33 and opp not on 42: 41 → 42
4. if opp on 21: 22 → 23
5. 41 → 42 (cand.)



Entry 50

1. if opp on 43 and opp not on 13: 32 → 33
2. if opp on 22 and opp not on 42: 41 → 42
3. if opp on 33 and opp not on 31: 41 → 31
4. if opp on 11 and opp on 13: 32 → 33
5. if opp on 34 and row 4 has ≥ 1 opp: 23 → 33

6. if opp on 21 and col 2 has ≥ 1 opp: 23 → 22
7. if row 3 empty of opp and row 4 empty of opp: 41 → 42

8. if opp on 11 and opp has no capturing move: 23 → 33

9. if opp on 34 and opp not on 31: 32 → 33

10. if opp not on 33 and col 4 empty of opp: 23 → 33

11. if opp not on 22 and col 2 has ≥ 1 opp: 23 → 22

12. if opp on 31 and opp not on 11: 41 → 42

13. if opp on 44 and opp not on 43: 41 → 42

14. if opp not on 11 and opp not on 21: 23 → 33

15. if opp on 11 and opp not on 14: 41 → 42

16. if opp on 11: 23 → 33

17. 32 → 42 (cand.)



Entry 51

1. if opp not on 23 and row 4 empty of opp: 41 → 42

2. 33 → 32 (cand.)

3. 33 → 43 (cand.)

4. 41 → 42 (cand.)



Entry 52

1. if opp on 43: 32 → 42

2. if opp not on 34 and opp has no capturing move: 33 → 43

3. if opp not on 31 and col 4 has ≥ 2 opp: 33 → 43

4. if opp on 42 and opp not on 31: 33 → 43

5. if col 2 has ≥ 1 opp and row 2 has ≥ 1 opp: 41 → 42

6. if col 3 has ≥ 1 opp and row 1 has ≥ 1 opp: 33 → 43

7. if opp on 31 and opp not on 42: 41 → 42

8. if opp on 34 and opp not on 23: 33 → 23

9. 32 → 42 (cand.)

10. 33 → 43 (cand.)



Entry 53

1. 33 → 43 (cand.)

2. 34 → 44 (cand.)



Entry 54

1. if opp not on 24 and col 3 empty of opp: 23 → 33

2. if col 2 empty of opp and row 2 empty of opp: 22 → 32

3. if opp on 12 and row 3 empty of opp: 42 → 32

4. if opp not on 31 and col 3 empty of opp: 23 → 33

5. if opp not on 11 and col 2 empty of opp: 22 → 32

6. if opp on 12 and opp on 31: 42 → 32
7. if opp not on 24 and opp not on 33: 23 → 33
8. if opp not on 22 and col 4 has ≥ 1 opp: 22 → 32
9. 42 → 32 (cand.)



Entry 55

1. if opp on 21 and opp not on 41: 42 → 41
2. if opp not on 23 and opp not on 33: 22 → 23
3. 22 → 32 (cand.)



Entry 56

1. if opp not on 23 and row 3 empty of opp: 31 → 32
2. if opp on 34 and opp not on 24: 23 → 33
3. if opp on 33: 42 → 32
4. if opp on 24 and opp not on 21: 31 → 32
5. 23 → 22 (cand.)



Entry 57

1. if col 3 empty of opp and row 1 has ≥ 1 opp: 32 → 33
2. if opp not on 12 and row 2 empty of opp: 22 → 23
3. if col 1 has ≥ 2 opp and row 3 empty of opp: 32 → 33
4. if opp not on 41 and row 4 has ≥ 1 opp: 42 → 41
5. if opp on 31 and opp not on 24: 42 → 43
6. if opp on 34: 32 → 33
7. if opp on 12 and row 2 has ≥ 1 opp: 42 → 43
8. if opp not on 23 and opp not on 33: 22 → 23
9. if opp not on 14 and opp not on 33: 32 → 33
10. if opp not on 11 and opp not on 14: 32 → 31
11. if opp on 11: 42 → 43
12. if opp on 14 and opp on 23: 42 → 43
13. if opp on 12: 22 → 23
14. 32 → 31 (cand.)



Entry 58

1. if col 3 empty of opp and row 2 empty of opp: 23 → 33
2. if row 2 empty of opp and row 3 empty of opp: 32 → 33
3. if opp on 33 and row 2 empty of opp: 42 → 43
4. if opp not on 43 and row 2 empty of opp: 32 → 33
5. if row 2 has ≥ 2 opp and row 1 empty of opp: 23 → 33
6. if opp on 24 and opp on 44: 42 → 43
7. if opp on 44 and opp has no capturing move: 32 → 33
8. if opp on 33 and opp not on 22: 32 → 22
9. if opp on 11 and col 3 has ≥ 1 opp: 32 → 33

10. if opp on 41 and col 3 has ≥ 1 opp: 32 → 33
11. if opp not on 34 and opp has a capturing move: 23 → 33
12. if opp not on 24 and col 1 has ≥ 2 opp: 32 → 22
13. if opp on 13 and row 1 has ≥ 2 opp: 32 → 33
14. if opp on 34 and opp not on 31: 23 → 33
15. if opp on 14 and row 4 has ≥ 1 opp: 32 → 33
16. if opp on 31 and col 2 has ≥ 1 opp: 23 → 33
17. if opp on 12 and opp not on 22: 32 → 22
18. if opp on 14 and opp not on 31: 42 → 43
19. if opp not on 34 and row 4 has ≥ 1 opp: 23 → 33
20. if opp on 24 and col 4 has ≥ 2 opp: 32 → 22
21. if opp not on 21 and opp not on 33: 32 → 33
22. if opp on 21 and opp on 24: 42 → 43
23. if opp on 12: 42 → 43
24. 32 → 33 (cand.)



Entry 59

1. if opp not on 43 and row 4 has ≥ 1 opp: 42 → 43
2. if opp on 43 and opp not on 41: 31 → 41
3. if opp on 21 and opp not on 41: 42 → 41
4. if opp not on 33 and col 1 has ≥ 1 opp: 32 → 33
5. if opp not on 33 and col 2 has ≥ 1 opp: 42 → 43
6. if opp on 33: 31 → 41
7. if opp on 13 and opp on 23: 32 → 33
8. 31 → 41 (cand.)



Entry 60

1. if opp on 23 and row 2 has ≥ 2 opp: 42 → 43
2. if opp on 21 and row 1 has ≥ 2 opp: 33 → 23
3. if opp on 24 and row 1 has ≥ 2 opp: 33 → 23
4. if opp on 12 and col 1 has ≥ 2 opp: 42 → 32
5. if opp on 21 and col 4 empty of opp: 33 → 23
6. if opp on 43 and row 4 has ≥ 2 opp: 22 → 23
7. if opp on 44 and col 2 has ≥ 1 opp: 33 → 43
8. if opp on 44 and row 3 has ≥ 1 opp: 22 → 32
9. if row 2 empty of opp and row 3 empty of opp: 22 → 23
10. if opp on 44 and opp not on 41: 33 → 43
11. if opp on 24 and col 3 has ≥ 1 opp: 22 → 23
12. if opp on 41 and col 4 empty of opp: 33 → 43
13. if opp not on 41 and row 2 has ≥ 2 opp: 42 → 32
14. if opp on 24 and opp not on 41: 33 → 23
15. if opp on 21 and row 4 empty of opp: 33 → 23
16. if opp on 21 and row 3 has ≥ 1 opp: 42 → 32
17. if col 1 empty of opp and col 2 empty of opp: 22 → 32
18. if opp on 23 and opp has a capturing move: 42 → 43
19. if opp on 43 and opp not on 32: 42 → 32

20. if opp not on 23 and col 3 has ≥ 1 opp: 22 $\rightarrow$ 23
21. if opp not on 32 and row 2 empty of opp: 42 $\rightarrow$ 32
22. if opp not on 13 and opp not on 41: 42 $\rightarrow$ 43
23. if opp not on 44 and row 3 empty of opp: 33 $\rightarrow$ 43
24. if opp on 24 and opp not on 31: 33 $\rightarrow$ 23
25. if opp not on 14 and opp not on 34: 22 $\rightarrow$ 32
26. if opp on 14 and opp not on 21: 33 $\rightarrow$ 43
27. if opp on 41 and opp not on 32: 33 $\rightarrow$ 23
28. 22 $\rightarrow$ 23 (cand.)



Entry 61

1. if opp on 22 and opp not on 12: 33 $\rightarrow$ 32
2. if row 2 empty of opp and row 4 empty of opp: 23 $\rightarrow$ 22
3. if opp on 31 and row 1 has ≥ 1 opp: 42 $\rightarrow$ 32
4. if col 4 has ≥ 2 opp and row 1 empty of opp: 23 $\rightarrow$ 22
5. if opp on 34 and col 4 has ≥ 2 opp: 42 $\rightarrow$ 32
6. if opp not on 24 and col 1 has ≥ 2 opp: 23 $\rightarrow$ 22
7. if opp not on 43 and row 4 has ≥ 1 opp: 33 $\rightarrow$ 43
8. if opp on 24 and opp not on 32: 33 $\rightarrow$ 32
9. if col 4 has ≥ 1 opp and col 2 empty of opp: 42 $\rightarrow$ 32
10. if opp on 12 and col 1 has ≥ 1 opp: 23 $\rightarrow$ 22
11. if opp not on 44 and row 3 empty of opp: 33 $\rightarrow$ 32
12. if opp on 43 and opp not on 34: 23 $\rightarrow$ 22
13. 42 $\rightarrow$ 32 (cand.)
14. 42 $\rightarrow$ 43 (cand.)



Entry 62

1. if opp on 43 and opp not on 41: 31 $\rightarrow$ 41
2. if opp on 44 and opp not on 43: 33 $\rightarrow$ 43
3. if opp on 22: 31 $\rightarrow$ 32
4. if col 2 empty of opp and row 2 empty of opp: 31 $\rightarrow$ 32
5. if opp on 23 and opp has a capturing move: 42 $\rightarrow$ 43
6. if opp on 21 and opp on 34: 31 $\rightarrow$ 32
7. if col 3 empty of opp and row 2 empty of opp: 33 $\rightarrow$ 43
8. if opp on 34 and opp has no capturing move: 31 $\rightarrow$ 32
9. if opp on 23 and opp not on 41: 31 $\rightarrow$ 41
10. if opp on 13 and opp has no capturing move: 33 $\rightarrow$ 43
11. if opp not on 24 and row 1 has ≥ 1 opp: 31 $\rightarrow$ 32
12. if opp on 41 and opp not on 43: 33 $\rightarrow$ 43
13. if opp on 24 and opp has no capturing move: 33 $\rightarrow$ 23

14. if opp on 13 and opp not on 11: 33 $\rightarrow$ 23
15. if opp on 41: 33 $\rightarrow$ 34
16. if opp on 11: 42 $\rightarrow$ 43
17. 31 $\rightarrow$ 32 (cand.)



Entry 63

1. if opp on 44 and opp not on 43: 33 $\rightarrow$ 43
2. if col 3 empty of opp and col 4 empty of opp: 33 $\rightarrow$ 43
3. if col 1 has ≥ 2 opp and col 4 empty of opp: 32 $\rightarrow$ 22
4. if row 1 empty of opp and row 4 empty of opp: 42 $\rightarrow$ 43
5. if opp on 23 and opp has a capturing move: 42 $\rightarrow$ 43
6. if opp not on 41 and row 1 empty of opp: 42 $\rightarrow$ 41
7. if opp on 31 and row 2 empty of opp: 33 $\rightarrow$ 23
8. if row 1 has ≥ 2 opp and row 2 empty of opp: 32 $\rightarrow$ 22
9. if opp on 24 and opp has a capturing move: 33 $\rightarrow$ 23
10. if opp not on 14 and col 1 has ≥ 2 opp: 32 $\rightarrow$ 22
11. if opp on 23 and col 2 empty of opp: 32 $\rightarrow$ 31
12. if opp on 24 and opp not on 31: 33 $\rightarrow$ 23
13. if opp on 41 and opp not on 43: 33 $\rightarrow$ 43
14. if opp on 43 and row 4 has ≥ 2 opp: 32 $\rightarrow$ 31
15. if opp on 12 and col 1 has ≥ 1 opp: 32 $\rightarrow$ 22
16. if opp on 21 and opp not on 11: 33 $\rightarrow$ 23
17. if opp not on 34 and col 3 empty of opp: 33 $\rightarrow$ 43
18. if opp on 22 and col 4 has ≥ 1 opp: 33 $\rightarrow$ 23
19. if opp on 34 and opp on 43: 42 $\rightarrow$ 41
20. if opp on 13 and opp has a capturing move: 33 $\rightarrow$ 34
21. if opp not on 22 and opp not on 34: 32 $\rightarrow$ 22
22. if opp not on 23: 32 $\rightarrow$ 31
23. 42 $\rightarrow$ 43 (cand.)



Entry 64

1. 34 $\rightarrow$ 33 (cand.)
2. 42 $\rightarrow$ 41 (cand.)
3. if opp on 33 and row 1 has ≥ 1 opp: 34 $\rightarrow$ 44
4. if opp on 33 and row 4 has ≥ 2 opp: 31 $\rightarrow$ 32
5. if opp on 33 and opp not on 22: 34 $\rightarrow$ 44
6. 31 $\rightarrow$ 32 (cand.)



Entry 65

1. if opp on 31 and opp not on 33: 34 $\rightarrow$ 33
2. if opp not on 22 and col 3 empty of opp: 34 $\rightarrow$ 33
3. if opp on 41 and opp has no capturing move: 32 $\rightarrow$ 31
4. if opp on 44 and opp not on 11: 42 $\rightarrow$ 43

5. if col 4 empty of opp and row 1 has ≥ 2 opp: 34 $\rightarrow$ 44
6. if opp on 14 and opp not on 41: 34 $\rightarrow$ 33
7. if col 4 has ≥ 1 opp and opp has no capturing move: 32 $\rightarrow$ 33
8. if opp not on 11 and opp not on 43: 34 $\rightarrow$ 44
9. if opp not on 13 and col 2 has ≥ 1 opp: 34 $\rightarrow$ 33
10. if opp on 11 and opp on 21: 32 $\rightarrow$ 33
11. if opp not on 12 and col 1 has ≥ 1 opp: 34 $\rightarrow$ 44
12. 34 $\rightarrow$ 33 (cand.)



Entry 66

1. if opp not on 43 and row 4 has ≥ 1 opp: 33 $\rightarrow$ 43
2. if opp not on 32 and opp has no capturing move: 34 $\rightarrow$ 44
3. if opp on 23 and opp not on 43: 42 $\rightarrow$ 43
4. if opp on 22: 33 $\rightarrow$ 32
5. if opp on 43: 34 $\rightarrow$ 44
6. if opp on 11 and opp on 24: 42 $\rightarrow$ 43
7. 34 $\rightarrow$ 44 (cand.)



Entry 67

1. if opp on 13 and opp not on 22: 42 $\rightarrow$ 32
2. if opp on 24 and opp not on 12: 42 $\rightarrow$ 43
3. if opp on 11: 42 $\rightarrow$ 32
4. 23 $\rightarrow$ 33 (cand.)



Entry 68

1. if opp on 33 and opp not on 31: 41 $\rightarrow$ 31
2. if opp on 14 and opp not on 43: 42 $\rightarrow$ 43
3. if opp on 43 and col 4 has ≥ 1 opp: 32 $\rightarrow$ 33
4. if opp on 44 and opp has no capturing move: 42 $\rightarrow$ 43
5. if opp on 21: 32 $\rightarrow$ 31
6. if col 4 has ≥ 1 opp and opp has no capturing move: 42 $\rightarrow$ 43
7. if opp not on 22 and opp not on 31: 32 $\rightarrow$ 33
8. if opp on 43 and opp not on 23: 32 $\rightarrow$ 33
9. if opp on 31: 42 $\rightarrow$ 43
10. if opp on 22 and opp on 34: 32 $\rightarrow$ 33
11. 41 $\rightarrow$ 31 (cand.)



Entry 69

1. if col 2 empty of opp and row 3 has ≥ 1 opp: 42 $\rightarrow$ 32
2. if col 2 empty of opp and opp has no capturing move: 41 $\rightarrow$ 31
3. if opp on 23 and opp not on 43: 42 $\rightarrow$ 43
4. if opp on 22 and row 4 has ≥ 1 opp: 33 $\rightarrow$ 32
5. if opp on 44 and opp not on 43: 42 $\rightarrow$ 43
6. if opp on 34 and opp not on 11: 42 $\rightarrow$ 32

7. if opp on 22 and opp not on 12: 33 $\rightarrow$ 32
8. if opp not on 31 and row 2 empty of opp: 41 $\rightarrow$ 31
9. if opp not on 43 and col 1 empty of opp: 42 $\rightarrow$ 43
10. if opp on 21: 42 $\rightarrow$ 43
11. if opp not on 13 and opp not on 31: 41 $\rightarrow$ 31
12. if opp on 12: 42 $\rightarrow$ 32
13. 33 $\rightarrow$ 23 (cand.)



Entry 70

1. if opp not on 12 and row 4 empty of opp: 43 $\rightarrow$ 33
2. if opp on 44 and opp not on 42: 41 $\rightarrow$ 42
3. if opp not on 11 and opp not on 13: 23 $\rightarrow$ 33
4. 43 $\rightarrow$ 33 (cand.)



Entry 71

1. 31 $\rightarrow$ 32 (cand.)



Entry 72

1. if opp on 34 and col 1 empty of opp: 43 $\rightarrow$ 33
2. if opp on 23 and row 4 has ≥ 1 opp: 32 $\rightarrow$ 33
3. if opp on 33 and col 1 empty of opp: 41 $\rightarrow$ 31
4. if opp on 23 and col 4 has ≥ 1 opp: 32 $\rightarrow$ 33
5. if opp on 34 and row 1 has ≥ 1 opp: 43 $\rightarrow$ 33
6. if opp on 44 and col 1 empty of opp: 32 $\rightarrow$ 33
7. if col 1 empty of opp and row 1 has ≥ 2 opp: 32 $\rightarrow$ 33
8. if opp on 14 and opp on 31: 43 $\rightarrow$ 33
9. if opp on 23 and col 2 has ≥ 1 opp: 32 $\rightarrow$ 33
10. if row 3 empty of opp and opp has a capturing move: 41 $\rightarrow$ 42
11. if opp on 14 and row 4 empty of opp: 43 $\rightarrow$ 44
12. if opp not on 31 and opp has a capturing move: 41 $\rightarrow$ 31
13. if opp on 44 and opp not on 42: 41 $\rightarrow$ 42
14. if opp on 31 and opp has no capturing move: 43 $\rightarrow$ 33
15. if col 3 has ≥ 1 opp and row 3 empty of opp: 43 $\rightarrow$ 44
16. if opp on 21 and row 1 empty of opp: 32 $\rightarrow$ 31
17. if opp on 42 and opp not on 11: 43 $\rightarrow$ 33
18. if opp on 22 and opp on 31: 41 $\rightarrow$ 42
19. if opp on 31 and opp not on 33: 43 $\rightarrow$ 33
20. if opp on 42 and opp not on 22: 41 $\rightarrow$ 31
21. if col 1 has ≥ 1 opp and col 2 has ≥ 1 opp: 43 $\rightarrow$ 44
22. 32 $\rightarrow$ 33 (cand.)
23. 41 $\rightarrow$ 31 (cand.)
24. if opp on 31 and opp not on 11: 43 $\rightarrow$ 42
25. 41 $\rightarrow$ 42 (cand.)



Entry 73

1. if col 2 empty of opp and row 3 empty of opp: 33 → 32
2. if opp on 42 and row 3 empty of opp: 33 → 32
3. if opp on 34 and col 2 empty of opp: 41 → 42
4. if opp on 44 and row 3 empty of opp: 33 → 34
5. if opp on 44 and opp not on 42: 41 → 42
6. if opp not on 32 and col 2 has ≥ 2 opp: 33 → 32
7. if opp on 32 and col 3 has ≥ 1 opp: 43 → 44
8. if opp on 12 and opp on 23: 41 → 42
9. if opp on 13 and row 2 empty of opp: 33 → 32
10. if opp on 24 and opp on 32: 33 → 34
11. if row 2 has ≥ 1 opp and row 1 empty of opp: 33 → 32
12. if opp on 14 and opp on 24: 33 → 23
13. if opp not on 32 and col 4 has ≥ 2 opp: 33 → 32
14. if col 2 has ≥ 2 opp and row 1 has ≥ 1 opp: 43 → 44
15. if opp on 12 and opp not on 21: 33 → 32
16. if opp on 34 and opp not on 42: 41 → 42
17. if opp on 22 and row 1 has ≥ 2 opp: 41 → 42
18. if opp on 11 and opp on 24: 41 → 42
19. if opp on 12 and opp not on 14: 43 → 42
20. if opp on 14 and opp on 22: 33 → 32
21. if col 3 empty of opp and row 1 has ≥ 1 opp: 43 → 44
22. if opp on 31 and opp not on 34: 33 → 32
23. if opp on 32: 41 → 31
24. if opp not on 11 and opp not on 24: 43 → 44
25. 33 → 23 (cand.)
26. 43 → 42 (cand.)



Entry 74

1. if opp on 23: 34 → 33
2. if opp on 32: 43 → 33
3. if opp on 44 and opp not on 42: 41 → 42
4. if opp on 14 and row 1 has ≥ 2 opp: 34 → 33
5. if opp not on 42 and opp has no capturing move: 41 → 42
6. 43 → 44 (cand.)
7. 41 → 31 (cand.)



Entry 75

1. if opp on 12 and opp not on 32: 42 → 32
2. if opp not on 41 and col 3 empty of opp: 43 → 33
3. if opp on 31 and opp has no capturing move: 42 → 32
4. if opp not on 33 and col 4 empty of opp: 43 → 33
5. if opp on 13 and opp on 41: 22 → 23
6. if opp on 14 and row 2 empty of opp: 43 → 33
7. if opp not on 33 and opp has a capturing move: 22 → 32
8. if opp on 34 and opp not on 33: 43 → 33

9. if opp on 21 and opp on 44: 43 → 33
10. if opp on 41 and row 3 empty of opp: 42 → 32
11. if opp on 21 and opp not on 24: 42 → 32
12. if opp on 13 and col 1 has ≥ 1 opp: 22 → 23
13. if opp not on 12 and opp not on 41: 43 → 33
14. if opp on 41 and opp on 44: 22 → 23
15. 42 → 32 (cand.)



Entry 76

1. if opp on 22: 31 → 32
2. if col 3 empty of opp and opp has no capturing move: 43 → 33
3. if opp on 33 and opp not on 34: 42 → 32
4. if opp on 21 and col 4 has ≥ 1 opp: 42 → 41
5. if opp not on 33 and row 2 empty of opp: 43 → 33
6. if col 4 empty of opp and opp has no capturing move: 43 → 44
7. if opp not on 21 and opp not on 23: 42 → 32
8. if opp not on 14 and opp not on 21: 43 → 33
9. if opp on 21: 42 → 41
10. 31 → 32 (cand.)



Entry 77

1. if col 3 empty of opp and row 3 has ≥ 1 opp: 43 → 33
2. if opp not on 33 and row 2 empty of opp: 43 → 33
3. if opp on 23 and col 4 has ≥ 1 opp: 32 → 33
4. if opp on 41 and col 3 empty of opp: 43 → 33
5. if opp not on 33 and row 3 has ≥ 1 opp: 43 → 33
6. if opp not on 12 and col 4 has ≥ 2 opp: 42 → 41
7. if opp not on 11 and col 3 empty of opp: 43 → 33
8. if opp on 23 and opp not on 13: 32 → 33
9. if opp on 13 and col 4 has ≥ 1 opp: 32 → 33
10. if opp on 13 and col 1 has ≥ 1 opp: 43 → 44
11. if opp not on 41 and row 2 empty of opp: 42 → 41
12. if col 4 has ≥ 1 opp and col 2 empty of opp: 32 → 31
13. if opp on 12 and col 4 empty of opp: 43 → 44
14. if opp on 22 and col 4 has ≥ 1 opp: 32 → 33
15. if opp not on 14 and opp not on 33: 43 → 33
16. if opp on 21 and opp on 33: 42 → 41
17. if opp on 33: 43 → 44
18. if opp on 11 and opp on 24: 43 → 33
19. 32 → 22 (cand.)



Entry 78

1. 31 → 32 (cand.)

Entry 79



1. if opp on 33 and opp not on 31: $41 \rightarrow 31$
2. if row 4 empty of opp and opp has no capturing move: $44 \rightarrow 43$
3. if opp on 42 and opp not on 31: $44 \rightarrow 43$
4. if opp on 22: $41 \rightarrow 42$
5. if opp on 43 and row 3 empty of opp: $32 \rightarrow 33$
6. if opp not on 34 and row 1 has ≥ 2 opp: $41 \rightarrow 42$
7. if opp on 42 and opp not on 34: $44 \rightarrow 43$
8. if opp on 31 and col 4 empty of opp: $41 \rightarrow 42$
9. if opp not on 21 and row 4 has ≥ 1 opp: $32 \rightarrow 33$
10. if opp on 21: $32 \rightarrow 31$
11. $44 \rightarrow 43$ (cand.)

References

- [1] E. Grädel, W. Thomas and T. Wilke (eds.). *Automata, Logics, and Infinite Games*. Lecture Notes in Computer Science 2500, Springer, 2002.
- [2] R. M. McConnell, K. Mehlhorn, S. Näher and P. Schweitzer. Certifying algorithms. *Computer Science Review* 5(2):119–161, 2011.
- [3] J. W. Romein and H. E. Bal. Solving Awari with parallel retrograde analysis. *IEEE Computer* 36(10):26–33, 2003.
- [4] T. Ströhlein. *Untersuchungen über kombinatorische Spiele*. Ph.D. thesis, Technische Hochschule München, 1970.
- [5] A. Tarski. A lattice-theoretical fixpoint theorem and its applications. *Pacific Journal of Mathematics* 5(2):285–309, 1955.